



QUALITY INFORMATION DOCUMENT

QUALITY INFORMATION DOCUMENT

Global Ocean Reanalysis Products GLOBAL_REANALYSIS_PHY_001_030

Issue: 1.6

Contributors: Marie Drévillon, Jean-Michel Lellouche, Charly Régner, Gilles Garric, Clément Bricaud, Olga Hernandez, Romain Bourdallé-Badie

Approval Date by Quality Assurance Review Group: 23/11/2023

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	Ref Date Issue	CMEMS-GLO-QUID-001-030 November 2023 1.6
--	----------------------	--

CHANGE RECORD

Issue	Date	§	Description of Change	Author	Checked By
1.0	18 January 2018	all	First version of document		
1.1	24 April 2018	title	Correct number of product in the document	Yann Drillet	
1.3	11 December 2020	NB	Add note to describe the switch to ERA5 forcing	Romain Bourdallé-Badie	
1.4	13/01/2021	all	Add dataset global-reanalysis-phy-001-030-monthly-climatology	Marie Drévillon	Marie Drévillon
1.5	03/12/2021	all	First temporal extension (covering 2020)	Romain Bourdallé-Badie	
1.6	18/10/2023		Add description of interim production	Romain Bourdallé-Badie	Marie Drévillon

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	Ref Date Issue	CMEMS-GLO-QUID-001-030 November 2023 1.6
--	-------------------------------	---

TABLE OF CONTENTS

I	EXECUTIVE SUMMARY	5
I.1	Products covered by this document	5
I.2	Summary of the results	6
I.2.1	Temperature.....	6
I.2.2	Salinity	7
I.2.3	Ocean currents	7
I.2.4	Sea level.....	7
I.2.5	Transport.....	8
I.2.6	Sea ice concentration	8
I.3	Estimated Accuracy Numbers.....	9
II	Production Subsystem description	10
II.1	GLORYS12V1 reanalysis (001_030)	10
III	Validation framework	15
IV	Validation results of GLORYS12V1	16
IV.1	Temperature	16
IV.1.1	T-CLASS1-T3D-MEAN	16
IV.1.2	T-CLASS1-SST-MEAN.....	17
IV.1.3	T-CLASS1-SST-TREND	18
IV.1.4	T-CLASS2-MOORINGS.....	19
IV.1.5	T-CLASS3-SST_QNET_MEAN.....	21
IV.1.6	T-CLASS3-HC_LAYER	21
IV.1.7	T-CLASS4-LAYER	22
IV.1.7.1	Timeseries	23
IV.1.7.2	2D Maps.....	24
IV.1.8	T-DA-INNO-STAT	26
IV.2	Salinity	27
IV.2.1	S-CLASS1-S3D_MEAN	27
IV.2.2	S-CLASS1-SSS-TREND.....	28
IV.2.3	S-CLASS2-MOORINGS	29
IV.2.4	S-CLASS3-SSS_SFW_2DMEAN	31
IV.2.5	S-CLASS3-SC_LAYER.....	32
IV.2.6	S-CLASS4-LAYER.....	33
IV.2.6.1	Timeseries	33
IV.2.6.2	2D Maps.....	34
IV.2.7	S-DA-INNO-STAT.....	36
IV.3	Currents	37
IV.3.1	UV-CLASS1-15m_MEAN.....	37
IV.3.2	UV-CLASS2-MOORINGS.....	38
IV.3.3	UV-CLASS2-MKE_1000m	40
IV.4	Sea level.....	40
IV.4.1	SL-CLASS3-2DMEAN	40
IV.4.2	SL-CLASS4	41
IV.4.3	SL-DA-INNO_STAT	41
IV.5	Transport	42
IV.5.1	UV-CLASS3-VOL_TRANSP.....	42
IV.5.2	UV-CLASS3-AMOC	43
IV.5.3	T-CLASS3-MHT.....	43

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	Ref Date Issue	CMEMS-GLO-QUID-001-030 November 2023 1.6
--	-------------------------------	---

IV.6 Sea Ice.....	44
IV.6.1 SI-CLASS1-CONC_MEAN	44
IV.6.2 SI-CLASS1-CONC_EXTREME	46
IV.6.3 SI-CLASS3-EXTENT.....	47
IV.6.4 SI-CLASS3-EXTENT_ANO.....	47
IV.6.5 SI-CLASS3-VOL	48
IV.6.6 SI-CLASS3-VOL_ANO.....	48
IV.6.7 SI-CONC-DA-INNO-STAT	49
 V System’s Noticeable events, outages or changes	 50
 VI Quality changes since previous version.....	 52
 VII References	 53

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	Ref CMEMS-GLO-QUID-001-030 Date November 2023 Issue 1.6
---	--

I EXECUTIVE SUMMARY

I.1 Products covered by this document

The product assessed in this document is referenced as:

GLOBAL-REANALYSIS-PHY-001-030 for Global Ocean Physics Reanalysis system product.

The goal of the Copernicus Marine Environment Monitoring Service (CMEMS) global ocean reanalysis is to provide an eddy resolving (1/12°) global ocean simulation, covering the recent period for which altimeter data is available (starting with the launch of TOPEX POSEIDON and ERS-1 satellites in the early 1990s), constrained by assimilation of observations and describing the space-time evolution of the following variables:

- * sea_surface_height_above_geoid
- * sea_water_potential_temperature
- * sea_water_salinity
- * sea_water_x_velocity
- * sea_water_y_velocity
- * sea_ice_area_fraction
- * sea_ice_thickness
- * sea_ice_x_velocity
- * sea_ice_y_velocity

These reanalyses are built to be as close as possible to the observations (i.e., realistic) and in agreement with the model physics. They cover the “altimetry era” (1st of January 1993 until 31 December 2022). The numerical products available to users are daily and monthly averages describing the ocean from surface to bottom (5900 m).

The GLOBALREANALYSISPHY-001-030 product consists of a global ocean reanalyses datasets at 1/12° horizontal resolution and is distributed on a regular 1/12° grid. See the Product User Manual (<https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-GLO-PUM-001-030.pdf>) for more details about the grid.

It is important to note that in the validation studies of this document, most of the time the bias is calculated as follows: "observations minus model". Hence, a model that is too warm model leads to negative values in the figures.

Table 1 summarizes the different data sets of the global ocean physical reanalysis products.

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	Ref Date Issue	CMEMS-GLO-QUID-001-030 November 2023 1.6
--	----------------------	--

Table 1: Overview of GLOBAL_REANALYSIS_PHY_001_030 product.

Product Name	Product origin	Production Unit	Data set	Kind of data set
GLOBAL_REANALYSIS_PHY_001_030	GLORYS12V1	GLO-MERCATOR-TOULOUSE-FR	global-reanalysis-phy-001-030-monthly-climatology-t global-reanalysis-phy-001-030-monthly-climatology-s global-reanalysis-phy-001-030-monthly-climatology-u-v global-reanalysis-phy-001-030-monthly-climatology-ssh global-reanalysis-phy-001-030-monthly-climatology-ice global-reanalysis-phy-001-030-monthly-t global-reanalysis-phy-001-030-monthly-s global-reanalysis-phy-001-030-monthly-u-v global-reanalysis-phy-001-030-monthly-ssh global-reanalysis-phy-001-030-monthly-ice dataset-global-reanalysis-phy-001-027-ran-fr-glorys12v1-coordinates global-reanalysis-phy-001-030-daily-t global-reanalysis-phy-001-030-daily-s global-reanalysis-phy-001-030-daily-u-v global-reanalysis-phy-001-030-daily-ssh global-reanalysis-phy-001-030-daily-ice	reanalysis

I.2 Summary of the results

I.2.1 Temperature

GLORYS12V1 reanalysis has weak regional biases (less than 0.4 °C) in temperature with respect to the World Ocean Atlas 2013 climatology and in-situ data (less than 0.1°C on global average with respect to in-situ temperatures). The largest biases occur in the [50 m-100 m] layer and in the northern Atlantic and Southern oceans.

The global mean sea surface temperature (SST) is close to observations with a weak (warm) bias of less than 0.1°C throughout the simulation period. The reanalysis also shows a linear trend of global SST increase which is consistent with AVHRR observations. The global mean surface net heat flux is weakly negative (- 0.7 W m⁻²).

Along the equator, temperature profiles are very consistent with observations with RMSD (Root mean square differences) generally smaller than 0.4°C throughout the water column, except for depths between 50 m-200 m where RMSD can reach 1.2°C in the Indian Ocean. Correlations with equatorial moorings also decrease with depth.

Heat content had been constantly increasing with time until 2011, when it started to stabilise. In contrast, deeper layers (below 2000 m) show a constant cooling.

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	Ref Date Issue	CMEMS-GLO-QUID-001-030 November 2023 1.6
---	-------------------------------	---

The thermal structure largely improves with the deployment of Argo buoys (after 2002), mainly in the upper layers (depth < 300 m) with a (cold) bias close to 0°C and RMSD with regard to in-situ observations of less than 1°C. RMSD decreases over time as the density of the observational network increases.

I.2.2 Salinity

Sea surface salinity (SSS) is generally fresher than the climatology with regional biases of less than 0.2 psu, except in the Arctic where values reach 2 psu. The global average of salinity bias with respect to salinity profiles is close to zero. Large fresh anomalies are found in tropical areas such as the western Atlantic and Pacific oceans. These surface fresh anomalies persist throughout the whole period covered by the reanalysis. In contrast, saltier surface waters are found in the Arctic Ocean (compared to the Levitus climatology). An overall salty bias (< 0.1 psu) is found in the [50 m-200 m] depth layer. Biases and RMSD significantly decrease over time in accordance with the increasing density of the observational network.

Trends of increasing SSS are seen over most parts of the global ocean. Yet, local negative trends are found in the Arctic Ocean and in the Indonesian throughflow. This overall weak (+0.002 per year) positive trend is reduced compared to the GLORYS2V4 reanalysis and mitigated by a negative trend during the first three years of the reanalysis.

Along the equator, salinity profiles are highly consistent with observations with RMSD generally below 0.2 psu throughout the water column, except near the surface where RMSD is higher and can reach 0.3 psu in the Atlantic Ocean. Correlations with equatorial moorings generally deteriorate with depth, although at the very surface, correlations are generally weaker than in the upper sub-surface layers.

The global salt content shows a weak positive trend with a stabilisation of sorts during the last decade. Large interannual variability is limited to the years prior to the full deployment of Argo buoys.

I.2.3 Ocean currents

GLORYS12V1 reanalysis reproduces the main ocean currents well. Along the boundary currents and Antarctic Circumpolar Current (ACC), mean kinetic energy (MKE) at 1000 m is higher than values in the ANDRO Argo drift data base. This possible overestimation of deep currents is still under investigation.

The simulated vertical dynamics and structure of the upper equatorial layers are in good agreement with moorings observations. In the Pacific, the easterly surface current strength is overestimated while currents in the core of the Equatorial Under Current (EUC) are underestimated. RMSD are generally smaller than 0.25 m s⁻¹ throughout the water column with a maximum in the lower part of the EUC. Correlations with equatorial moorings tend to deteriorate with depth.

I.2.4 Sea level

GLORYS12V1 reanalysis closely matches altimetry observations and has a good skill to describe sea level variability. Regional trends are particularly well reproduced. However, with 2.66 mm/year, the globally averaged increase in sea level is slightly underestimated compared to the Aviso SLA CCI estimates of 3.27 mm/year over this period (<https://www.aviso.altimetry.fr/en/data/products/ocean-indicators-products/mean-sea-level.html>). Altimetry observations at high northern latitudes are scarce, an area where strong sea level trends are depicted in GLORYS12V1. The globally averaged RMSD of the analysed sea level is about 5.5 cm while the forecast sea level is 0.5 cm higher. The overall global bias for the 2005-2016 time period is 0.07 cm.

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	Ref Date Issue	CMEMS-GLO-QUID-001-030 November 2023 1.6
---	-------------------------------	---

I.2.5 *Transport*

Compared to estimates by Lumpkin and Speer (2007), GLORYS12V1 generally overestimates the mean volume transport in all basins and particularly all along the ACC current (by up to 30 Sv). However, ACC volume transports at Drake passage simulated by GLORYS12V1 and GLORYS2V4 are in better agreement with latest estimations from Donohue et al. (2016) and Colin de Verdière and Ollitrault (2016). Four GLORYS12V1 transports in the northern Atlantic and Pacific ocean sections have opposite signs compared to Lumpkin and Speer (2007) estimates; however, they remain within the uncertainty estimates in Lumpkins and Speer (2007).

The Atlantic Meridional Overturning Circulation (AMOC) shows no significant trend throughout the reanalysis period. While it is 3 Sv weaker than estimates based on RAPID data (Moat et al., 2020), its correlation with RAPID data is high (0.86).

The meridional heat transport generally agrees with estimates by Ganachaud & Wunsch (2000) and Trenberth & Caron (2001) (within the given uncertainties).

I.2.6 *Sea ice concentration*

As a result of sea ice concentration assimilation, sea ice extent variability (seasonal cycles, interannual variability, and trends) is well reproduced in GLORYS12V1. Arctic sea ice extent and volume both show respective general decreases of -85,400 km²/year and -46,200 km³/year for the 1993-2016 period. Winter sea ice extent maxima in both hemispheres are overestimated compared to the assimilated CERSAT data. Although the year 2016 exhibits an unusually strong negative anomaly, overall, the Antarctic sea ice extent still exhibits a positive trend of 56,300 km²/year for the 1993-2016 period. RMSD and biases with regard to the assimilated ice concentration dataset are stable throughout the period. In the Arctic, the summer and winter maxima of RMSD with respect to the assimilated sea ice concentration dataset reach 0.25 and 0.12, respectively. If an independent dataset is used for the comparison, these results are only slightly higher although mostly unchanged. There is a positive bias between observations and model during spring and summer, due to an overestimation of ice melting in GLORYS12V1. During winter, the bias is also positive as the model underestimates the spread of ice extent. Results are similar in Antarctic, although with higher RMSD (up to 0.4).

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	Ref Date Issue	CMEMS-GLO-QUID-001-030 November 2023 1.6
--	----------------------	--

I.3 Estimated Accuracy Numbers

GLORYS12V1 1993-2016

Estimated accuracy numbers (EANs) are calculated using RMSD based on model/observation comparisons made by the data assimilation system.

Temperature (°C)

Level	Period	1993-2016		1993-2005		2005-2016	
		Mean	RMSD	Mean	RMSD	Mean	RMSD
0-100		-0.05	0.97	-0.03	1.00	-0.07	0.91
100-300		-0.02	0.85	0.01	0.93	-0.04	0.80
300-800		-0.005	0.48	0.01	0.55	-0.01	0.43
800-2000		0.006	0.20	-0.01	0.27	0.008	0.18

Salinity (psu)

Level	Period	1993-2016		1993-2005		2005-2015	
		Mean	RMSD	Mean	RMSD	Mean	RMSD
0-100		-0.003	0.293	-0.007	0.407	-0.002	0.26
100-300		-0.007	0.165	-0.015	0.249	-0.004	0.13
300-800		-0.001	0.082	0.002	0.112	-0.001	0.07
800-2000		0.000	0.054	0.003	0.091	-0.001	0.04

Sea level (cm)

Period	1993-2016		1993-2005		2005-2016	
	Mean	RMSD	Mean	RMSD	Mean	RMSD
	0.003	5.5	0.002	5.3	0.07	5.5

SST (°C)

Period	1993-2016		1993-2005		2005-2016	
	Mean	RMSD	Mean	RMSD	Mean	RMSD
	-0.05	0.5	-0.05	0.5	-0.05	0.5

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	Ref Date Issue	CMEMS-GLO-QUID-001-030 November 2023 1.6
---	-------------------------------	---

II PRODUCTION SUBSYSTEM DESCRIPTION

Production center: Mercator Ocean International, Toulouse, France (Global MFC)

Production system name: GLORYS12V1 (Global Ocean Reanalysis Products

GLOBAL_REANALYSIS_PHY_001_030)

II.1 GLORYS12V1 reanalysis (001_030)

Over the past years, Mercator Ocean has been regularly upgrading its global ocean physical reanalysis through improvements in the ocean model, assimilation scheme and assimilated data sets. The last upgrade concerned the eddy-permitting reanalysis GLORYS2V4 GLOBAL_REANALYSIS_PHY_001_025 ($\frac{1}{4}^\circ$ horizontal resolution and 75 vertical levels, <http://marine.copernicus.eu/documents/QUID/CMEMS-GLO-QUID-001-025.pdf>) covering the altimetry era (1993-2016). In the framework of the Copernicus Marine Environment Monitoring Service (CMEMS), we propose now an eddy-resolving physical reanalysis called GLORYS12V1 GLOBAL_ANALYSIS_FORECAST_001_030 (Lellouche et al, 2021), covering the same time period and based on the current real-time global forecasting CMEMS PSY4V3R1 system GLOBAL_ANALYSIS_FORECAST_001_024 ($1/12^\circ$ horizontal resolution and 50 vertical levels, <http://marine.copernicus.eu/documents/QUID/CMEMS-GLO-QUID-001-024.pdf>).

The physical model component is the NEMO platform (v3.1) driven at the surface by ECMWF ERA-Interim reanalysis until end of 2018 and ERA5 reanalysis thereafter¹. Observations are assimilated by means of a reduced-order Kalman filter with a 3D multivariate modal decomposition of the background error. This includes an adaptive-error estimate and localization algorithm. Along-track altimeter data (sea level anomaly – SLA), satellite sea surface temperature (SST), sea ice concentration, and in-situ vertical profiles of temperature and salinity (T/S) are jointly assimilated. Moreover, a 3D-VAR scheme provides a correction for the slowly-evolving large-scale biases in temperature and salinity.

Compared to PSY4V3R1 (see Lellouche et al, 2018), GLORYS12V1 reanalysis uses the reprocessed atmospheric forcing from the global atmospheric reanalysis ERA-Interim and benefits from a few changes in the system settings regarding observation errors.

Compared to GLORYS2V4, GLORYS12V1 reanalysis includes the following major updates: initial T/S conditions derived from EN4.2.0 data base, global steric effect added to model sea level, new Glacial Isostatic Adjustment correction, new seasonal observation error for assimilation of in-situ T/S vertical profiles, adaptive tuning of observational SLA and SST errors, additional Quality Control of the assimilated T/S vertical profiles based on dynamical height criteria, assimilation in the deep ocean (below 2000 m) of climatological T/S vertical profiles using a non-Gaussian error at depth. New and improved reprocessed in situ and satellite data sets have also been used.

In order to provide a synthetic view of the consistency between the PSY4V3R1 real time $1/12^\circ$ analyses GLOBAL_ANALYSIS_FORECAST_001_024 and the GLORYS12V1 reanalysis GLOBAL_ANALYSIS_FORECAST_001_030, a summary of a comparison between GLORYS12V1's configuration and the configurations used in PSY4V3R1 and GLORYS2V4 is provided in Tables 2 of this document. Complementary references and initial descriptions of the system set up can be found in <http://marine.copernicus.eu/documents/QUID/CMEMS-GLO-QUID-001-024.pdf> and <http://marine.copernicus.eu/documents/QUID/CMEMS-GLO-QUID-001-025.pdf>.

Sensitivity tests were performed for the year 2017 using GLORYS2V4 (Mercator Ocean global $\frac{1}{4}$ reanalysis, members of GREP product; see global-reanalysis-phy-001-031 CMEMS product). The switch in atmospheric forcing from ERAinterim to ERA5 with the upgrade to GLORYS12V1 led to no

¹ In early 2019, the atmospheric forcing flux in GLORYS12V1 reanalysis was changed from ERAinterim to ERA5.

significant changes in the EANS of oceanic variables, except for a reduction in sea ice volume and extent in the Southern Ocean during austral summer (Figure 0.1).

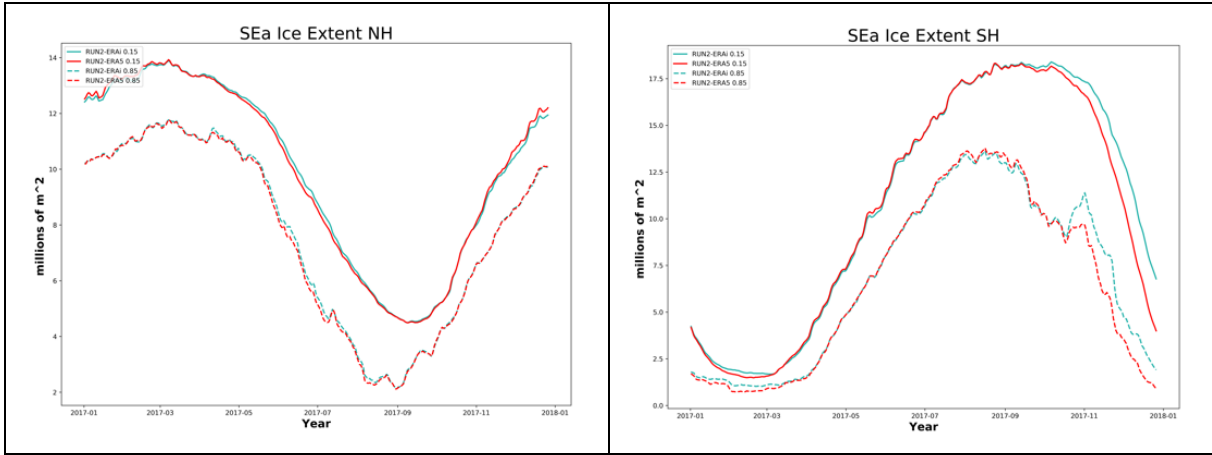


Figure 0.1 Evolution of sea ice extent for northern (left) and southern hemisphere (right) for the simulation forced with ERAinterim (blue) and ERA5 (red). To estimate the sea ice extend two criteria have been chosen: 85% (solid line) and 15% (dashed line).

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	Ref Date Issue	CMEMS-GLO-QUID-001-030 November 2023 1.6
--	----------------------	--

Table 2: Comparing model configurations for GLORYS12V1, PSY4V3R1, and GLORYS2V4.

PSY4V3R1 GLOBAL_ANALYSIS_FORECAST_001_024	GLORYS2V4 GLOBAL_REANALYSIS_PHY_001_025	GLORYS12V1 GLOBAL_REANALYSIS_PHY_001_030
50 z-levels	75 z-levels	identical to PSY4V3R1
ECMWF operational atmospheric analyses with 3H frequency No snowfall	ECMWF atmospheric reanalysis ERAinterim 3H/24H frequency + diurnal cycle Snowfall ERA5 after January 2019	identical to GLORYS2V4
Large scale correction of ECMWF operational atmospheric analyses precipitations towards PMWC (Hilburn, K. A., (2009))	Large scale correction of ERAi precipitations (towards GPCPV2.2 observations: https://climatedataguide.ucar.edu/climate-data/gpcp-monthly-global-precipitation-climatology-project) and radiative fluxes (towards GEWEX SRB3.0 & 3.1 observations: https://www.gewex.org/resources/data-sets/) ERA5 without correction after 2019.	Large scale correction of ERAi precipitations (towards PMWC observations) and radiative fluxes (towards GEWEX SRB3.0 & 3.1 observations) ERA5 without correction after 2019.
Initial Conditions (December 2006) estimated from EN4.2.0 monthly gridded fields for December 2006 using a regression technique	Initial Conditions (December 1991) estimated from EN4.0.2 monthly gridded fields for December 1991 using a regression technique	Initial conditions (December 1991) estimated from EN4.2.0 monthly gridded fields for December 1991 using a regression technique
Surface mass Budget EMP (Evaporation – Precipitation – Runoff)=0 + seasonal cycle + 2.2 mm/yr (2007-present).	Surface mass Budget EMP=0 + seasonal cycle + 1.74 mm/yr (1993-2016).	Surface mass Budget EMP=0 + seasonal cycle + 1.31 mm/yr (1993-2001) + 2.2 mm/yr (2002-2016).
Global steric effect diagnosed from T and S and added to SSH	No addition of global steric effect to SSH	identical to PSY4V3R1
Turbulent kinetic energy (TKE) mixing scheme with mixing length set to 30 m	TKE with mixing length set to 10 m	identical to GLORYS2V4
Restoring to WOA13 2005-2013 decadal climatological temperature and salinity at Gibraltar and Bab-El-Mandeb straits	Restoring to EN4.0.2 decadal climatology at Gibraltar and Bab-El-Mandeb straits	Restoring to EN4.2.0 decadal climatology at Gibraltar and Bab-El-Mandeb straits
Climatological Runoff Dai et al (2009) + freshwater fluxes from icebergs for Greenland and Antarctica	identical to PSY4V3R1	identical to PSY4V3R1
Solar penetration 3 bands (RGB)	Solar penetration (2 bands)	identical to GLORYS2V4

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	Ref Date Issue	CMEMS-GLO-QUID-001-030 November 2023 1.6
--	----------------------	--

Table 2 (cont.): Comparing data assimilation configurations of GLORYS12V1, PSY4V3R1, and GLORYS2V4.

PSY4V3R1 GLOBAL_ANALYSIS_FORECAST_001_024	G2V4 GLOBAL_REANALYSIS_PHY_001_025	G12V1 GLOBAL_REANALYSIS_PHY_001_030
Relative wind with 50% ocean currents in momentum flux	identical to PSY4V3R1	identical to PSY4V3R1
Outputs: High Frequency (1h) + daily mean + coarsened files at ¼° for biogeochemistry	Outputs : daily means	identical to PSY4V3R1
Assimilation of CORA4.1 in situ dataset until 2003 (early version of INSITU_GLO_TS_REP_OBSERVATIONS_013_001_b). From 2014 to present, the near-real time CMEMS product INSITU_GLO_NRT_OBSERVATIONS_013_030 is assimilated.	identical to PSY4V3R1	Assimilation of CORA4.1 until 2003. From 2014 to 2015, CORA5.0 product is assimilated. For 2016, CORA5.1 product is assimilated.
Assimilation of sea ice concentration from OSISAF SEAICE_GLO_SEAICE_L4_REP_OBSERVATIONS_011_009	Assimilation of sea ice concentration from CERSAT	identical to GLORYS2V4 + OSISAF only as verification
Assimilation of SST OSTIA SST_GLO_SST_L4_NRT_OBSERVATIONS_010_001	Assimilation of NOAA SST analyses at ¼° (AVHRR)	identical to GLORYS2V4
Hybrid Mean Dynamic Topography (MDT) based on CNES-CLS13 + new glacial isostatic adjustment file	Hybrid MDT based on CNES-CLS13	identical to PSY4V3R1
Error covariance (ocean + sea ice state anomalies) from twin simulation of PSY4V3R1 (only corrected from large scale T and S biases) + spatial filtering	Error covariance (ocean + sea ice state anomalies) from a consistent free global simulation	identical to PSY4V3R1
3D T/S in-situ observations errors from GLORYS2V3	3D T/S in-situ observation errors from GLORYS2V2	3D T/S in-situ seasonal observations errors from PSY4V3R1
Adaptive tuning for SST and SLA observation errors	Fixed errors from 2 cm to 5.5 cm (instrument dependency) for SLA and 0.6°C for SST + representativity error map	Adaptive tuning for SST and SLA observation errors + specification of a 2D minimum error map
2 layers of quality control are applied (QC1 & QC2)	Only QC1 is applied	identical to PSY4V3R1
Assimilation of WOA13 profiles below 2000 m with non-Gaussian observation error distribution	Restoring to Gouretski and Koltermann (2004) climatology below 2000 m and poleward of 60°S	Assimilation of EN4.2.0 profiles below 2000 m with non-Gaussian observation error distribution
3D-VAR bias correction on a 3-month window	identical to PSY4V3R1	3D-VAR bias correction on a 3-month window until the end of 2003, using a 1-month window thereafter

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	Ref Date Issue	CMEMS-GLO-QUID-001-030 November 2023 1.6
--	----------------------	--

Table 2 (cont): Comparing data assimilation configurations of GLORYS12V1, PSY4V3R1, and GLORYS2V4.

PSY4V3R1 GLOBAL_ANALYSIS_FORECAST_001_024	G2V4 GLOBAL_REANALYSIS_PHY_001_025	G12V1 GLOBAL_REANALYSIS_PHY_001_030
Total SSH increment from the analysis	SSH increment split in HBAR from the analysis and HDYN from 3D T/S increments	identical to PSY4V3R1
Global mean of total SSH increment = 0	Global mean of steric SSH increment = 0	identical to PSY4V3R1
TSUV equatorial pseudo-observations in [150 m-3000 m] layer	identical to PSY4V3R1	identical to PSY4V3R1
Due to adaptive tuning of observation errors, no amplification of SLA and SST observation error with observations density	Increase in SLA and SST observational error with increase in observation density	identical to PSY4V3R1
SST observation: one point out of 4 on the ¼° ORCA grid is assimilated	SST observation: one point out of 2 points on the ¼° ORCA grid is assimilated	identical to PSY4V3R1

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	Ref Date Issue	CMEMS-GLO-QUID-001-030 November 2023 1.6
---	-------------------------------	---

III VALIDATION FRAMEWORK

The validation method used here is described in Drevillon et al. (2018) and consists of a series of diagnostics that are used by all producers of global ocean reanalysis/reference forced simulations.

The reference period for the validation is 1 January 1993 – 31 December 2016. The times series are based on monthly fields except for the innovation diagnostics (see specific sections on T-DA_INNO_STAT, S-DA_INNO_STAT, SL-DA-INNO_STAT, SI-CONC-DA-INNO_STAT).

IV VALIDATION RESULTS OF GLORYS12V1

IV.1 Temperature

IV.1.1 T-CLASS1-T3D-MEAN

GLORYS12V1 reanalysis has weak regional biases (less than 0.4 °C) in temperature with respect to the World Ocean Atlas 2013 climatology (Fig. 1.1). The largest biases occur in the [50 m-100 m] layer and in the northern Atlantic and Southern oceans.

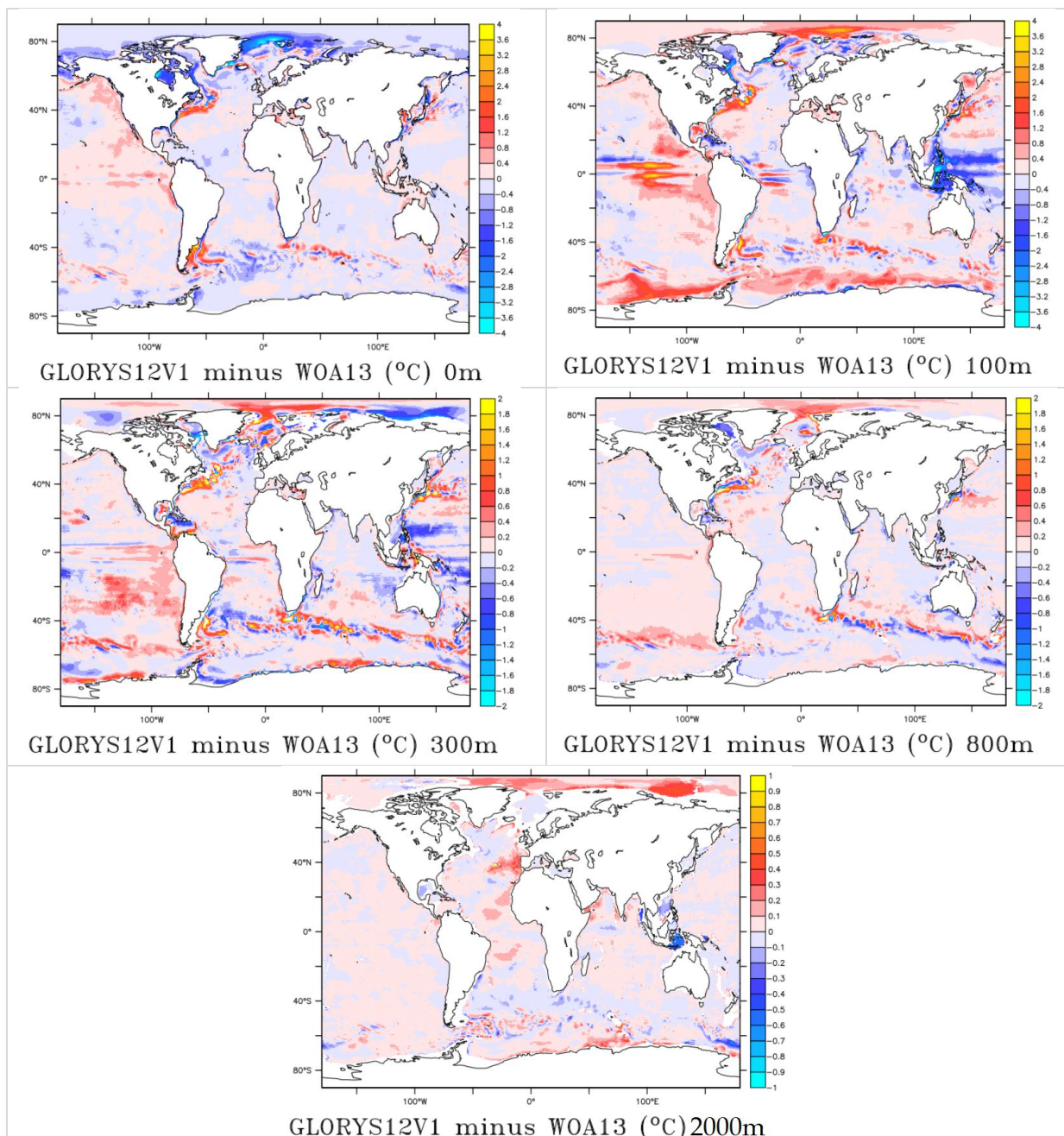


Fig. 1.1: Bias of temperature (°C) between GLORYS12V1 and the World Ocean Atlas 2013 climatology (WOA13) at different depths ranging from 0 to 2000 m. For this comparison, model data have been averaged over the period 1993 – 2016 (note colour scale changes between plots).

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	<table> <tr> <td>Ref</td><td>CMEMS-GLO-QUID-001-030</td></tr> <tr> <td>Date</td><td>November 2023</td></tr> <tr> <td>Issue</td><td>1.6</td></tr> </table>	Ref	CMEMS-GLO-QUID-001-030	Date	November 2023	Issue	1.6
Ref	CMEMS-GLO-QUID-001-030						
Date	November 2023						
Issue	1.6						

IV.1.2 T-CLASS1-SST-MEAN

The global mean SST is close to the observations with a weak (warm) bias of less than 0.1°C over the entire reanalysis period (Fig. 1.2).

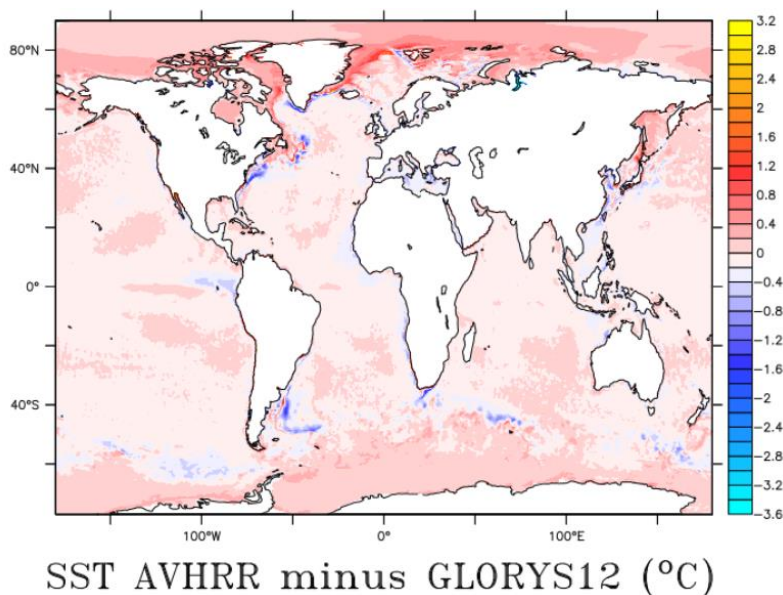


Fig. 1.2: Mean 1993-2016 difference in sea surface temperature (°C) between GLORYS12V1 and AVHRR data.

IV.1.3 T-CLASS1-SST-TREND

The positive linear trend in global SST (Fig 1.3) is highly consistent with AVHRR data.

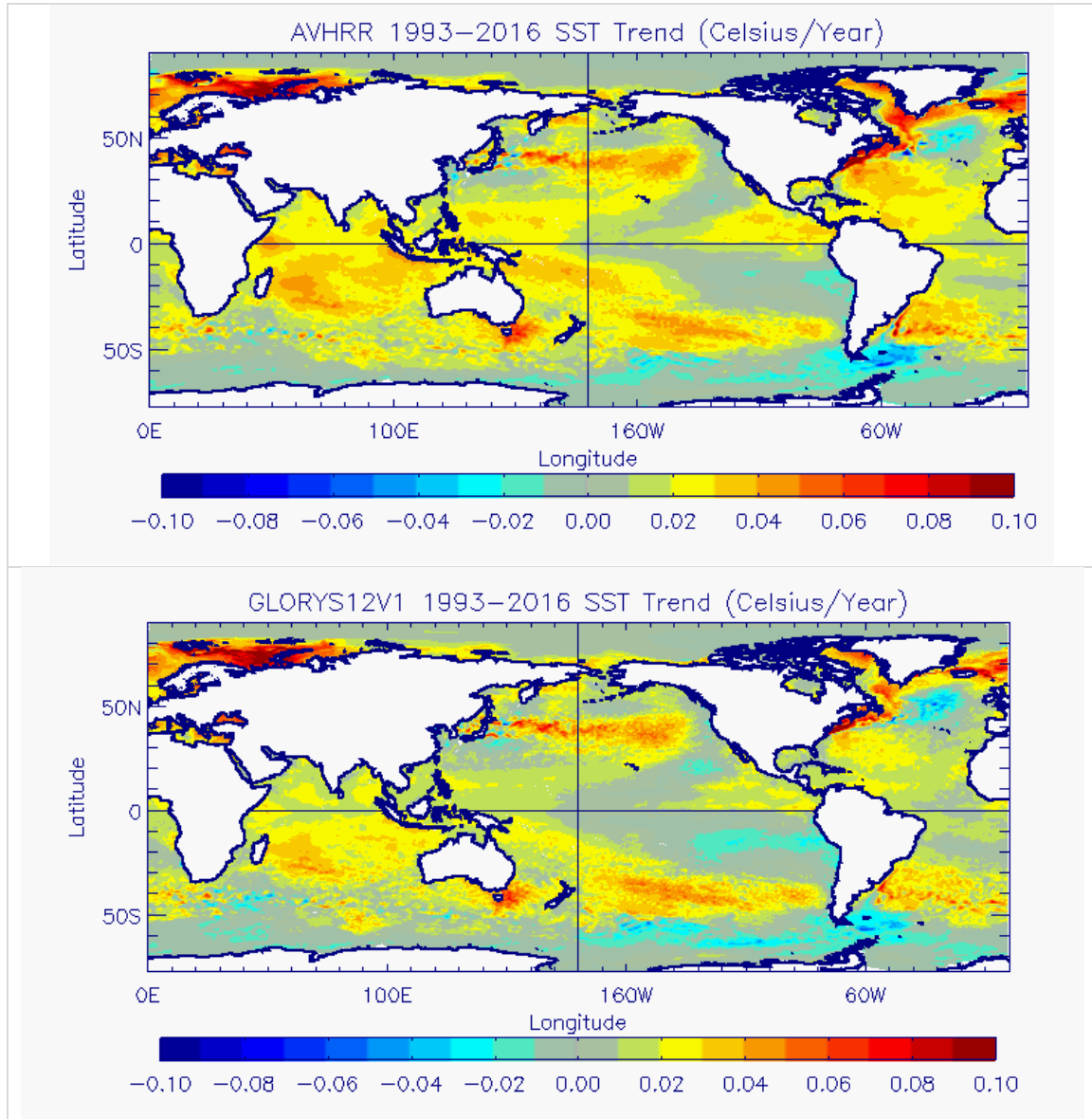


Fig. 1.3: Mean linear SST trend over years 1993–2016 in AVHRR data (upper panel) and GLORYS12V1 (bottom panel).

IV.1.4 T-CLASS2-MOORINGS

Along the equator, temperature profiles are very consistent with observations (Fig 1.4) with RMSD generally below 0.4°C throughout the water column, except for depths between 50 m-200 m where RMSD can reach 1.2°C in the Indian Ocean. Correlations with equatorial moorings deteriorate with depth.



Fig. 1.4: (left column) Multi-annual means (for years 1993-2015) of temperature profiles at the equator. (middle) RMSD and (right) correlations showing observations (black) and GLORYS12V1 (green). Mooring observations are from TAO/TRITON: 0N-140W, 0N-110W, 0N-165E, 0N-147E, PIRATA: 0N-23W, 0N-0E and RAMA: 0N-90E (continues on next page).

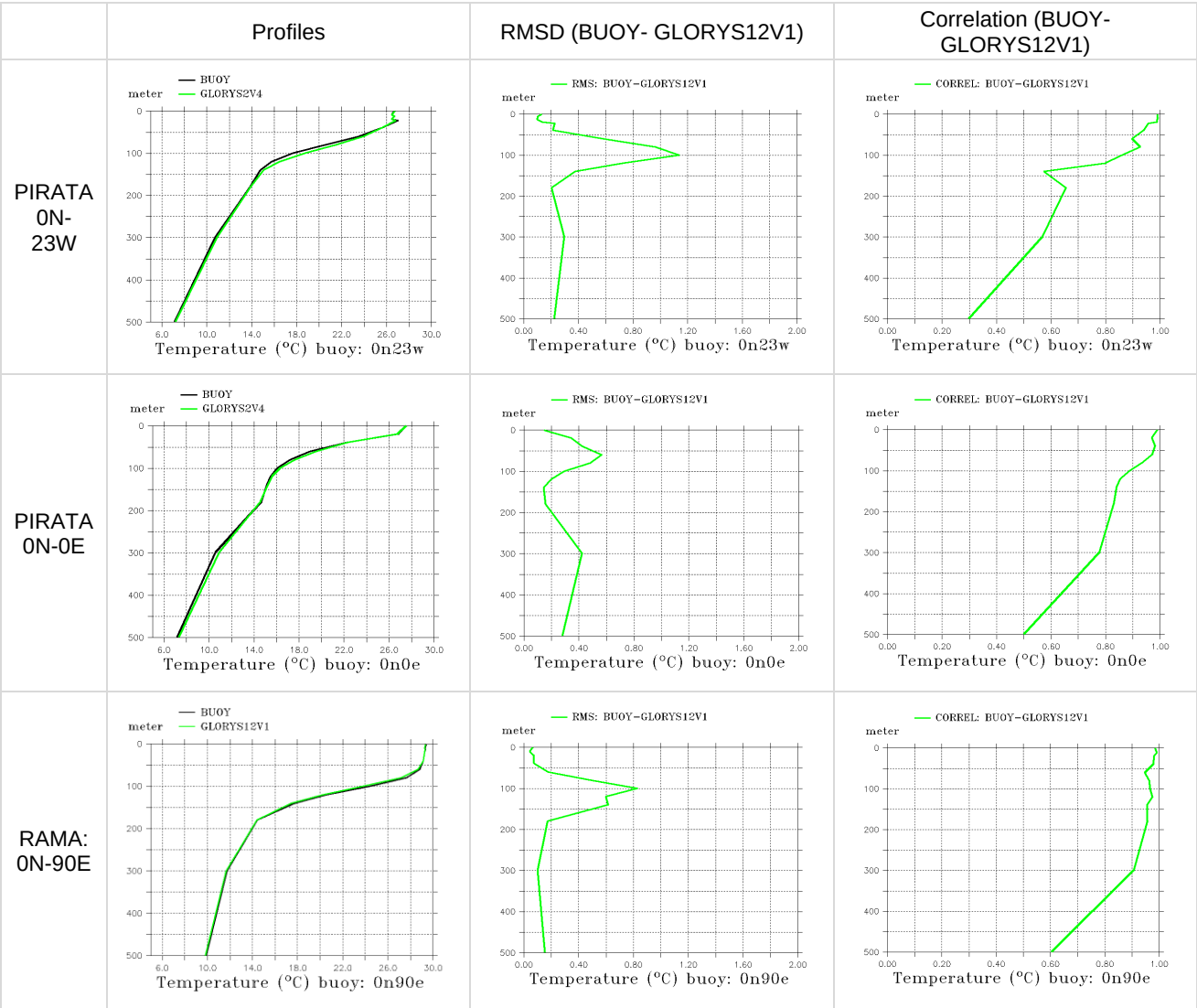


Fig. 1.4 (cont.): (left) Multi-annual means (for years 1993-2015 of temperature profiles at the equator. (middle) RMSD and (right) correlations showing observations (black) and GLORYS12V1 (green). Mooring observations are from TAO/TRITON: 0N-140W, 0N-110W,0N-165E,0N-147E, PIRATA:0N-23W,0N-0E and RAMA: 0N-90E.

IV.1.5 T-CLASS3-SST_QNET_MEAN

The global mean surface net heat flux is weakly negative over the 1993-2016 period (-0.7 W.m^{-2}) (Fig 1.5). The globally averaged linear trend for net surface heat flux is $-0.35 \text{ W m}^{-2} \text{ year}^{-1}$, while the global average linear SST trend is close to observations ($+0.013 \text{ }^{\circ}\text{C/year}$ in AVHRR vs $+0.018$ in GLORYS12V1).

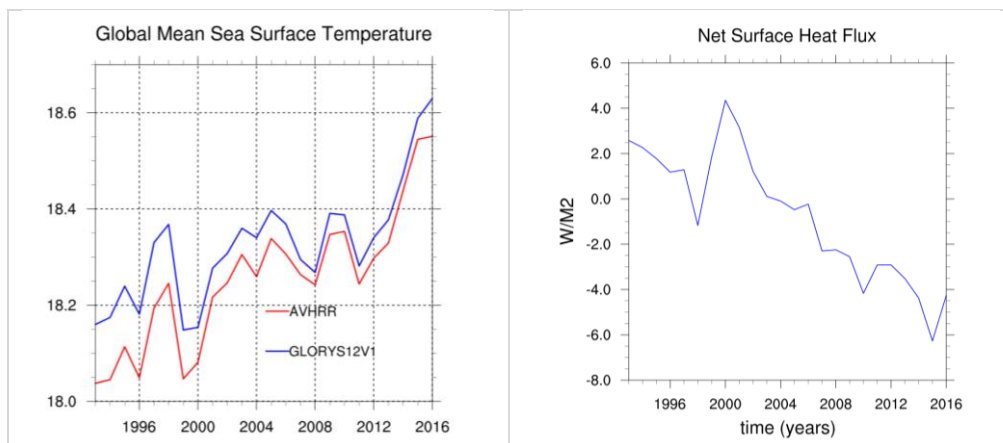


Fig. 1.5: Annually and globally averaged SST (left, $^{\circ}\text{C}$) and surface net heat flux (right, W.m^{-2}) for the 1993-2016 period. Assimilated SST AVHRR observations are in red (left panel only) and GLORYS12V1 in blue

IV.1.6 T-CLASS3-HC_LAYER

The heat content (fig 1.6) constantly increases with time until 2011. Since 2011, the heat content may have stabilised. Only the deeper layers (below 2000 m depth) show a constant cooling.

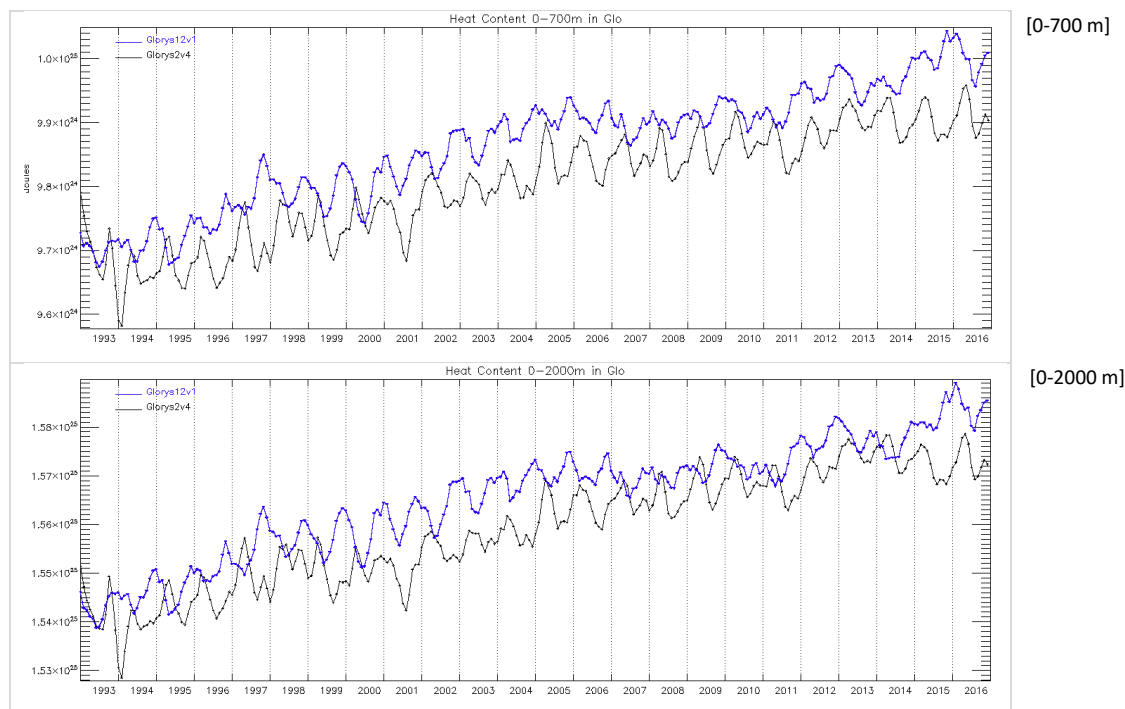


Fig. 1.6: Time series (1992-2016) of global mean temperature in different layers for GLORYS2V4 (black) and GLORYS12V1 (blue).

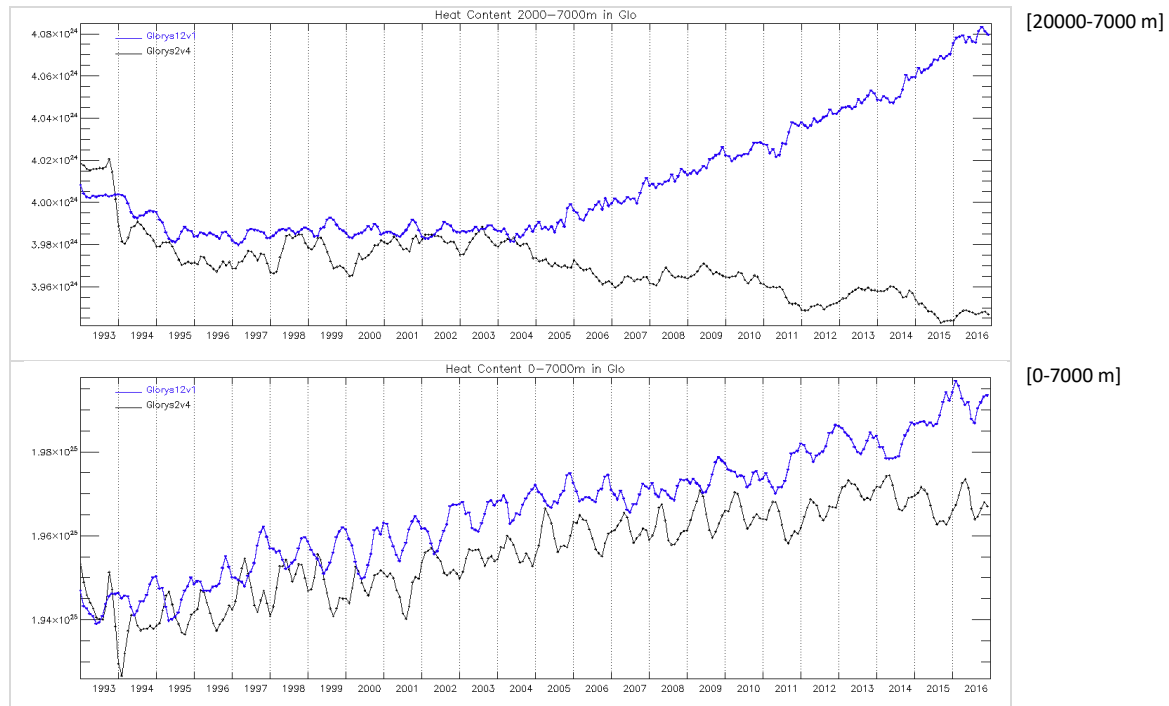


Fig. 1.6 (cont.): Time series (1992-2016) of global mean temperature in different layers for GLORYS2V4 (black) and GLORYS12V1 (blue).

IV.1.7 T-CLASS4-LAYER

For this validation we used the NSITU_GLO_TS_OA_NRT_OBSERVATIONS_013_002_A in-situ dataset from Copernicus Marine Service, performing daily subsampling prior to data assimilation.

RMSD in the global average timeseries looks very noisy until the appearance of first Argo floats in the 2000's (Fig 1.7.1). The number of in-situ data prior to the ARGO program is low with a poor spatial coverage (see also Fig 1.7.2). As a result, the global statistics are not very robust during the early years of the reanalysis period. With the emergence of ARGO data, the metrics stabilize for the two layers displayed in Figure 1.7.1: 0-50 m and 0-500 m. When we look at the differences between the model and the observations we see that the reanalysis is very stable from the 2000's up to 2016. For the temperature we have a small warm bias (up to 0.1°C) in the 0-500 m layer (located between 100 and 200 m). For that variable the reanalysis beat the climatology for the bias and the RMS with lower RMS differences in the 0-500 m layer depth.

IV.1.7.1 Timeseries

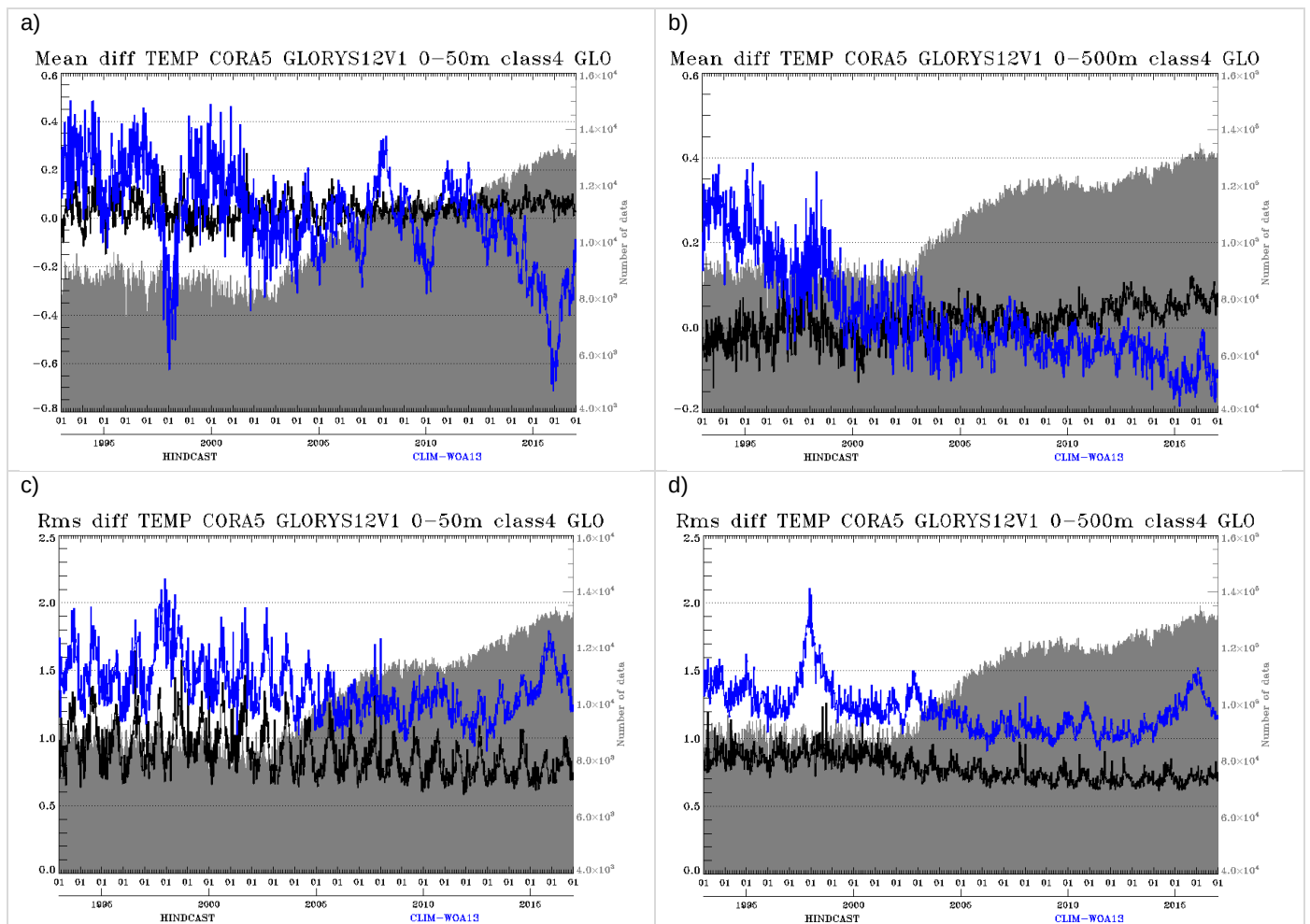


Figure 1.7.1 Timeseries (1993-2016) of (a,b) temperature bias and (c,d) RMSD between observed and model temperatures (in black) and between WOA13 climatologies and observations (in blue), in (a,c) the 0-50 m layer and (b,d) the 0-500 m layer. The grey shading on these graphics represents the number of data (cf., right y-axis for values)

The warm bias diagnosed in the timeseries (Fig 1.7.1) is clearly visible in the Gulf Stream region, the Yellow Sea, along the west coast in Africa, and to some extent in the Antarctic counter current region (Fig. 1.7.2).

IV.1.7.2 2D Maps

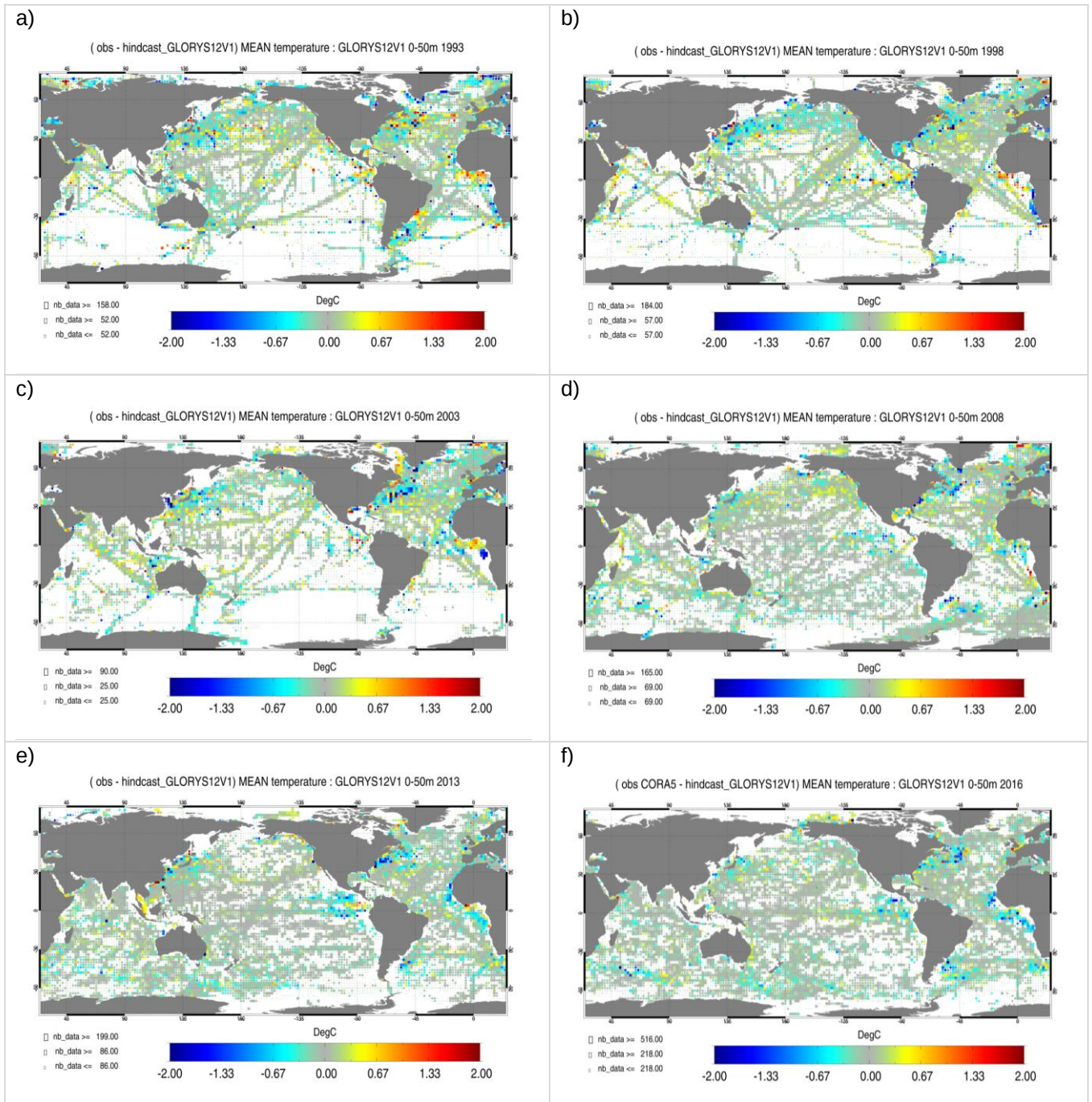


Figure 1.7.2: Bias temperature difference (observations-model) between the daily model averages and all available T/S observations from the Cora5.0 product for the years (a) 1993, (b) 1998, (c) 2003, (d) 2008, (e) 2013, and (f) 2016. Model outputs have been collocated with in situ sampling locations. Data are averaged over the top 50m and the size of each pixel is proportional to the number of observations used to compute the difference in 2°x2° grid boxes.

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	<table> <tr> <td>Ref</td><td>CMEMS-GLO-QUID-001-030</td></tr> <tr> <td>Date</td><td>November 2023</td></tr> <tr> <td>Issue</td><td>1.6</td></tr> </table>	Ref	CMEMS-GLO-QUID-001-030	Date	November 2023	Issue	1.6
Ref	CMEMS-GLO-QUID-001-030						
Date	November 2023						
Issue	1.6						

If we average over the top 500 m, the warm bias appears in the Pacific, the Gulf Stream region, and in the tropics (Fig 1.7.3). This warm bias is probably due to some issues in the mean sea surface height in these regions. The increase in data points over the years highlights the increased availability of in-situ observations. By 2016, the distribution of the in-situ data is fairly uniform all over the world, except in polar regions.

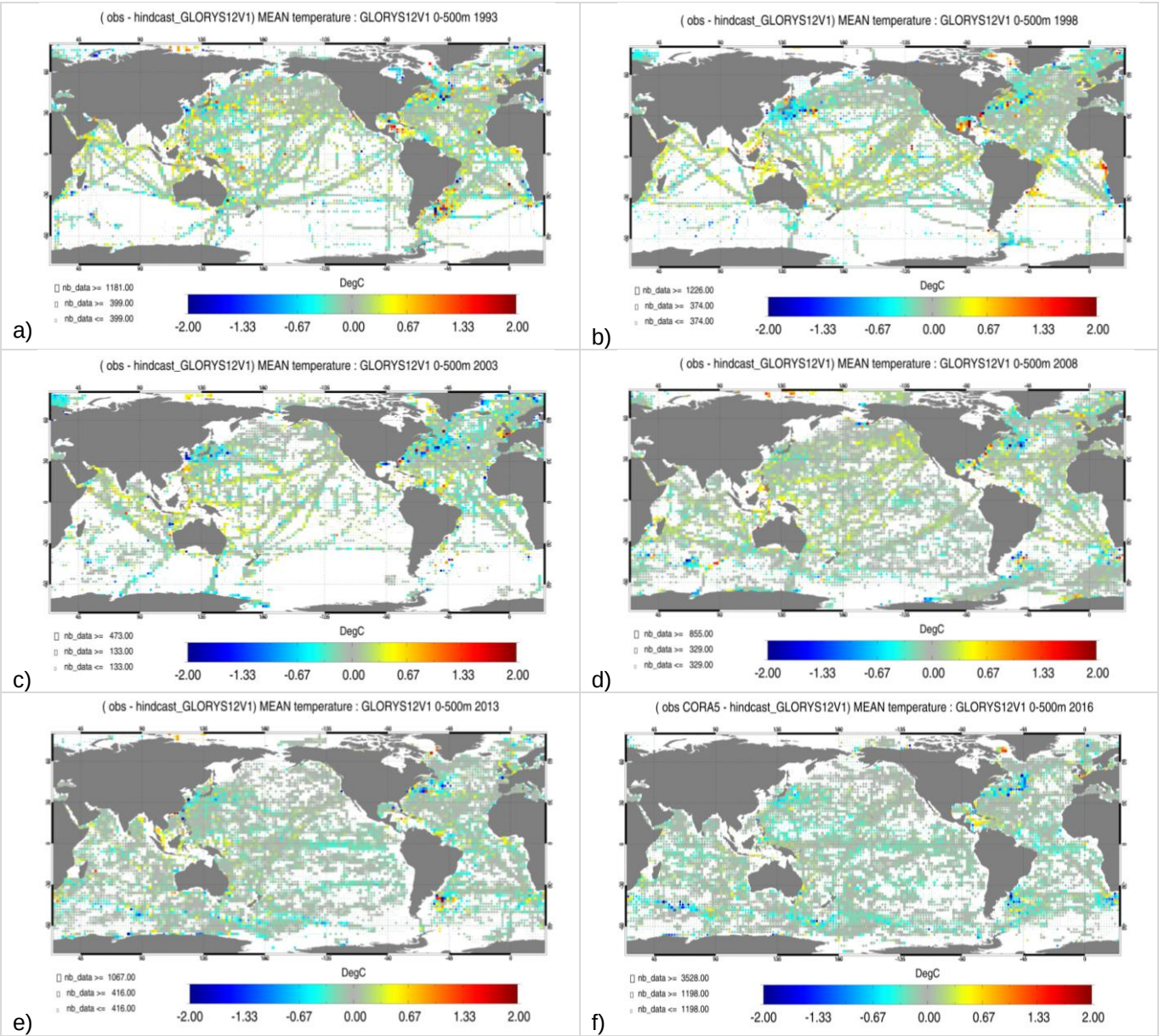


Figure 1.7.3: As in Figure 1.7.2 but for the 0-500 m layer.

IV.1.8 T-DA-INNO-STAT

The temperature innovations in Fig. 1.8 are based on the same assimilation statistics presented in Section IV.1.7 but here spatially integrated. A warm bias of the model is diagnosed at the surface (0.1°C on average) and a slight warming trend can be seen throughout the water column (observation – model differences) in Fig. 1.8. A focus on recent years shows that despite the fact that the average innovation stays stable in time, the seasonal bias linked to re-stratification issues in the model tends to increase at the end of the period.

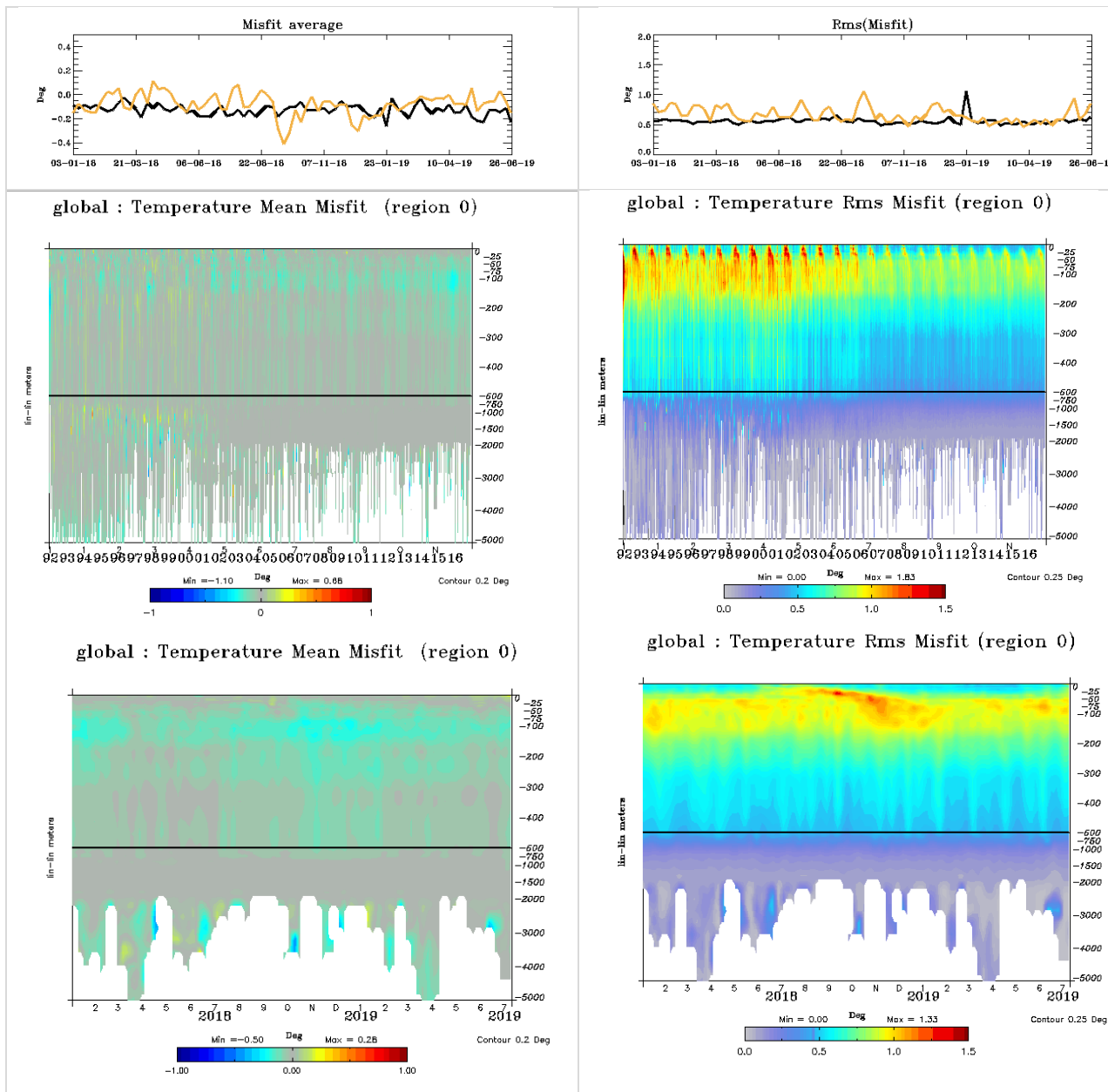


Fig. 1.8: Time series of the domain-averaged innovation (=region0 in the plot title) in terms of bias (left column) and RMSD (right column); (top) SST with AVHRR data in black and CORA4.1 SST data in orange ($^{\circ}\text{C}$), CORA 4.1 in-situ temperature profiles for the period: 1993 – 2016 (middle) and zoom on the more recent period Jan 2018- July 2019 (bottom).

IV.2 Salinity

IV.2.1 S-CLASS1-S3D_MEAN

Sea surface salinity (SSS) is generally fresher than the climatology with regional biases of less than 0.2 psu, except in the Arctic where larger biases appear (Fig. 2.1). Modelled salinities are also fresher than the climatology in the Amazon plume, around Indonesia, and in the Western Tropical Pacific where large precipitation events take place which may be overestimated. At mid latitudes, the model salinity is too high. As expected, the largest signals correspond to high-variability areas (strong currents, Indonesian throughflow).

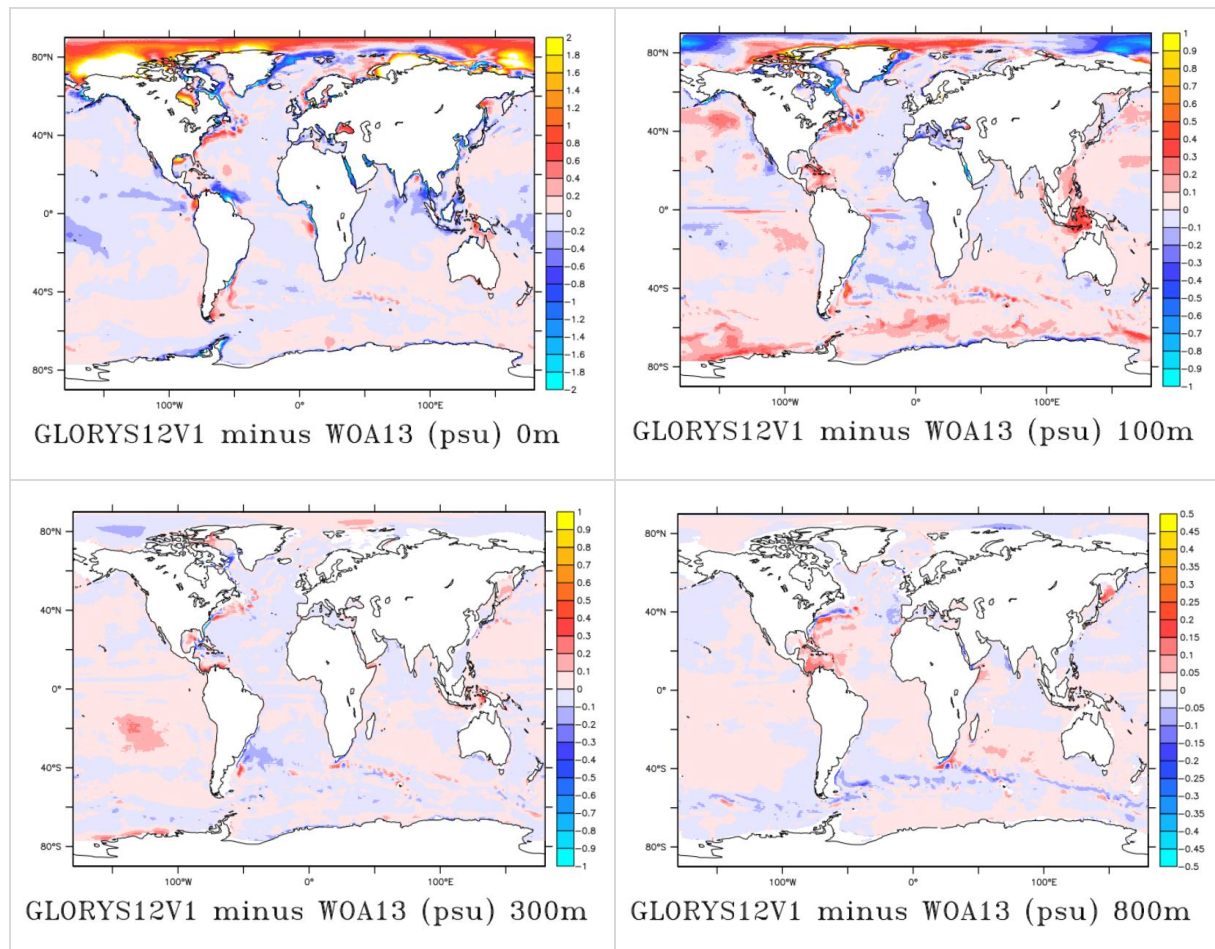


Fig. 2.1: BIAS in salinity (in psu) between GLORYS12V1 minus the World Ocean Atlas 2013 climatology (WOA13) near 0 m, 100 m, 300 m, 800 m and 2000 m depth. GLORYS12V1 data have been averaged over the period 1993 – 2016 (note colour scale changes).

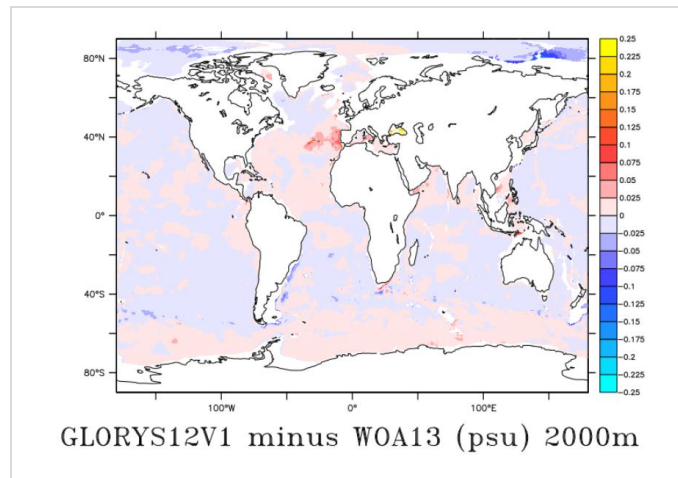


Fig. 2.1: (cont.) BIAS in salinity (in psu) between GLORYS12V1 minus the World Ocean Atlas 2013 climatology (WOA13) near 0 m, 100 m, 300 m, 800 m and 2000 m depth. GLORYS12V1 data have been averaged over the period 1993 – 2016 (note colour scale changes).

IV.2.2 S-CLASS1-SSS-TREND

Positive trends of SSS are seen over most parts of the global ocean (Fig 2.2). Yet, local negative trends are found in the Arctic Ocean and in the Indonesian throughflow. This overall weak (+0.002 per year) positive trend is reduced compared to the GLORYS2V4 reanalysis and mitigated by a negative trend during the first three years of the reanalysis.

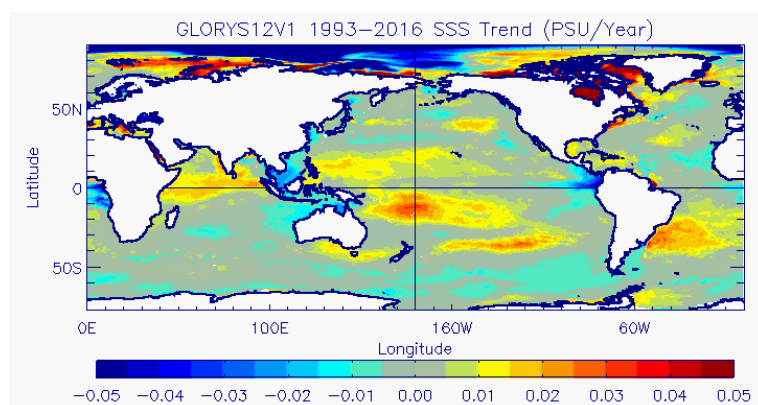


Fig. 2.2: Linear 1993-2016 SSS trend in GLORYS12V1.

IV.2.3 S-CLASS2-MOORINGS

Along the equator, salinity profiles (Fig. 2.3.1 and Fig. 2.3.2) are highly consistent with moorings' observations with RMSD generally smaller than 0.2 psu in the water column, apart at the surface where the RMSD is higher and can intermittently reach 0.3 psu in the Atlantic Ocean. Correlations with equatorial moorings also decrease with depth with however a generally weaker correlation at the surface compared to sub-surface scores.

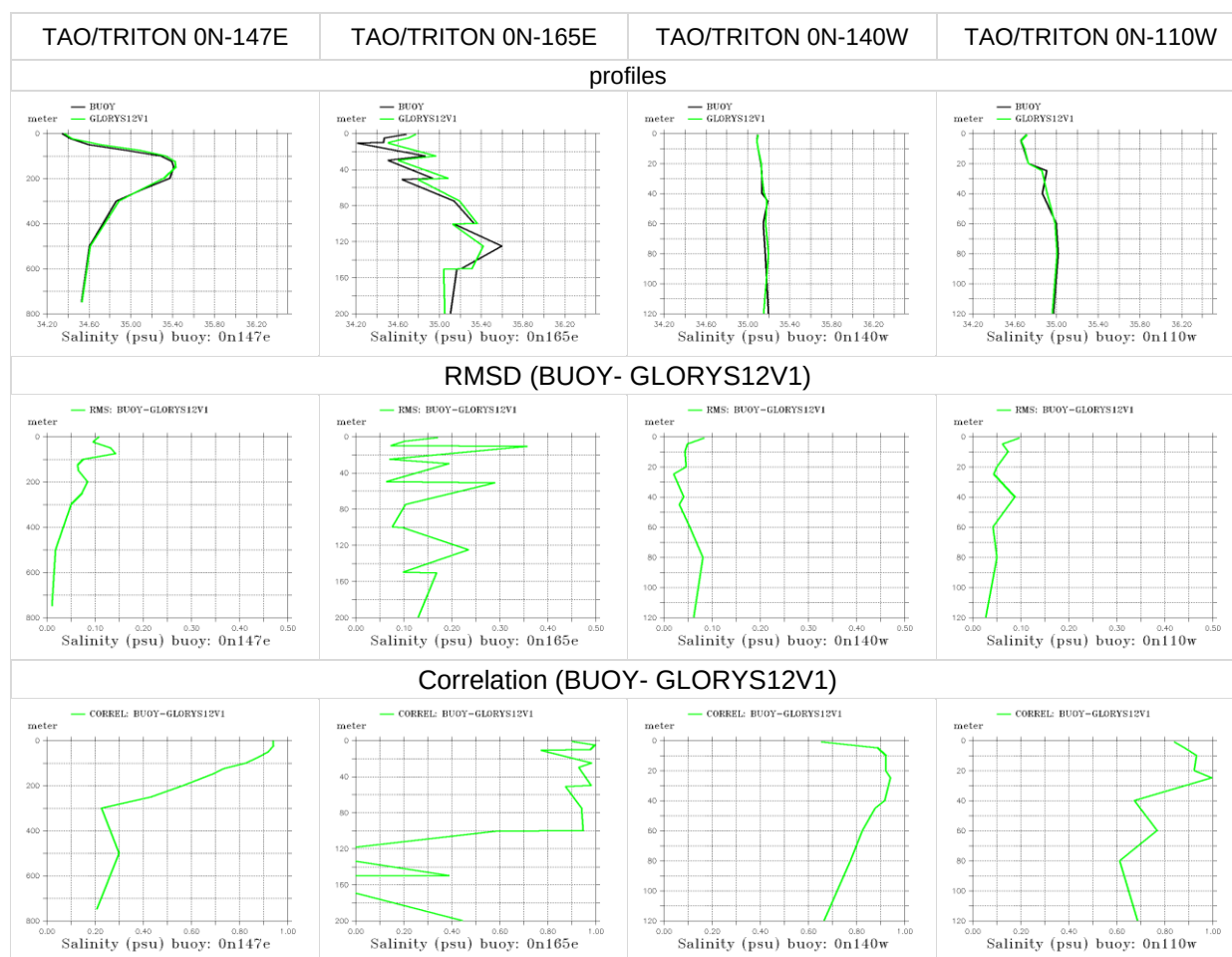


Fig. 2.3.1: Mean equatorial salinity profiles, (middle) RMSD, and (bottom) correlations estimated over the period 1993 – 2015. Observations (TAO/TRITON: 0N-140W, 0N-110W, 0N-165E, 0N-147E) are shown in black and GLORYS12V1 in green (continues in next page).

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	Ref Date Issue	CMEMS-GLO-QUID-001-030 November 2023 1.6
--	---	---

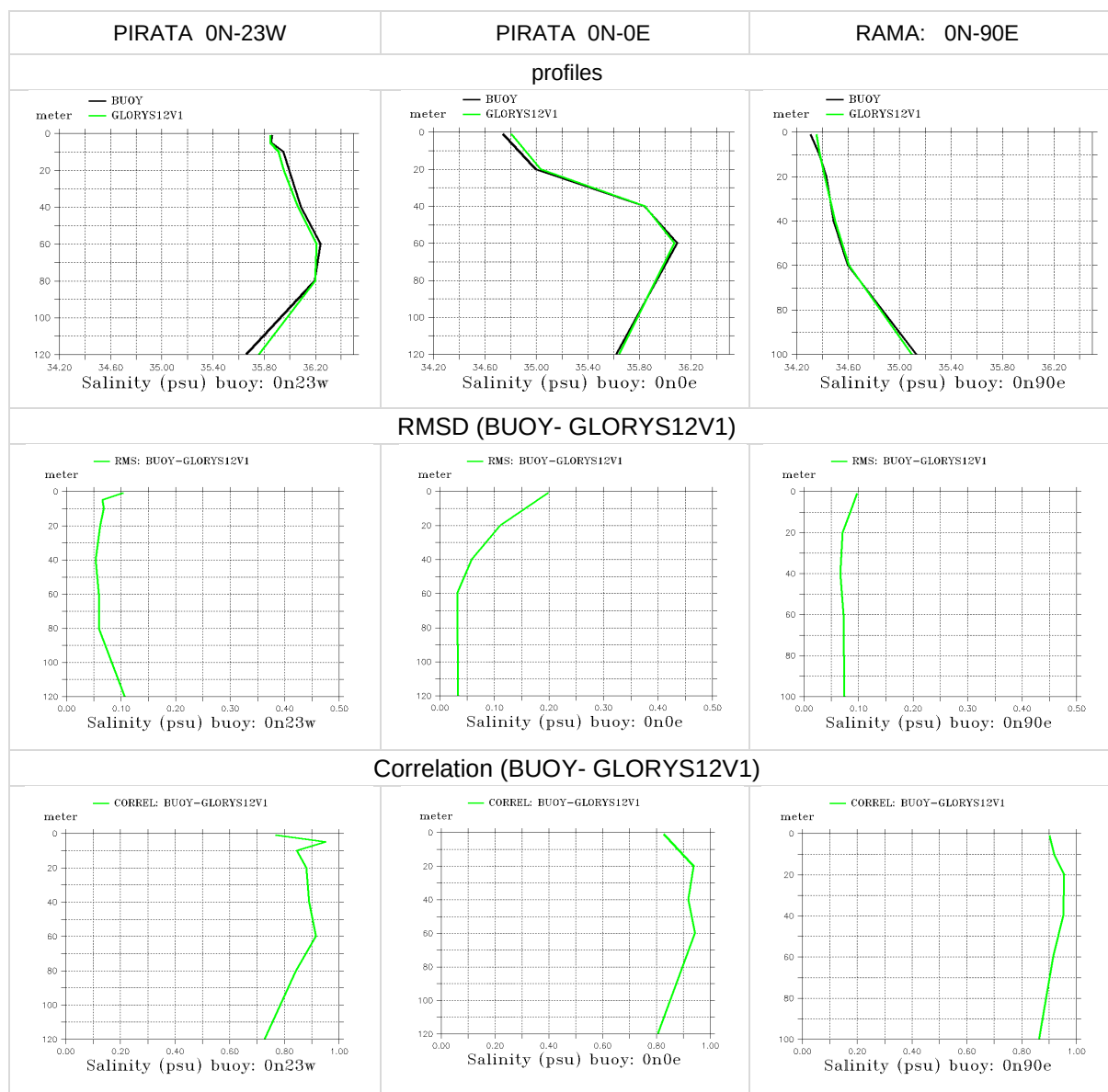


Fig. 2.3.2: Mean equatorial salinity profiles, (middle) RMSD, and (bottom) correlations estimated over the period 1993 – 2015. Observations (TAO/TRITON: 0N-140W, 0N-110W, 0N-165E, 0N-147E, PIRATA: 0N-23W, 0N-0E and RAMA: 0N-90E) are shown in black and GLORYS12V1 in green.

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	<table> <tr> <td>Ref</td><td>CMEMS-GLO-QUID-001-030</td></tr> <tr> <td>Date</td><td>November 2023</td></tr> <tr> <td>Issue</td><td>1.6</td></tr> </table>	Ref	CMEMS-GLO-QUID-001-030	Date	November 2023	Issue	1.6
Ref	CMEMS-GLO-QUID-001-030						
Date	November 2023						
Issue	1.6						

IV.2.4 S-CLASS3-SSS_SFW_2DMEAN

The long term mean value of the SSS is 34.52 for GLORYS12V1 and 34.51 for GLORYS2V4 and net surface fresh water input in both GLORYS12V1 and GLORYS2V4 is -0.005 mm/day (Fig. 2.4).

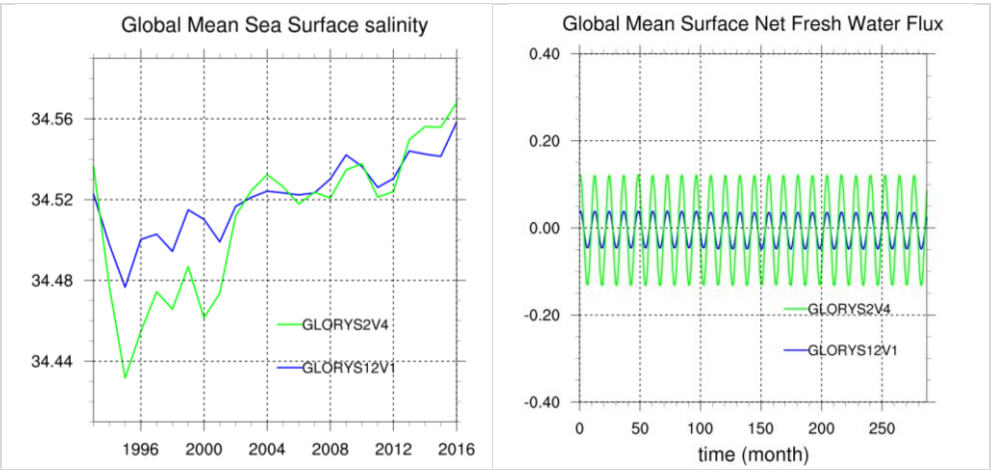


Fig. 2.4: Global spatial averages of annual mean SSS from (left,psu) and monthly mean surface net fresh water heat fluxes (right, mm day⁻¹) for the period 1993 – 2016 from GLORYS12V1 (blue) and GLORYS2V4 (green).

IV.2.5 S-CLASS3-SC_LAYER

The global salt content (Fig. 2.5) shows a weak positive trend that is stabilisation during the last decade. Large interannual variability is found prior the full deployment of the Argo buoys.

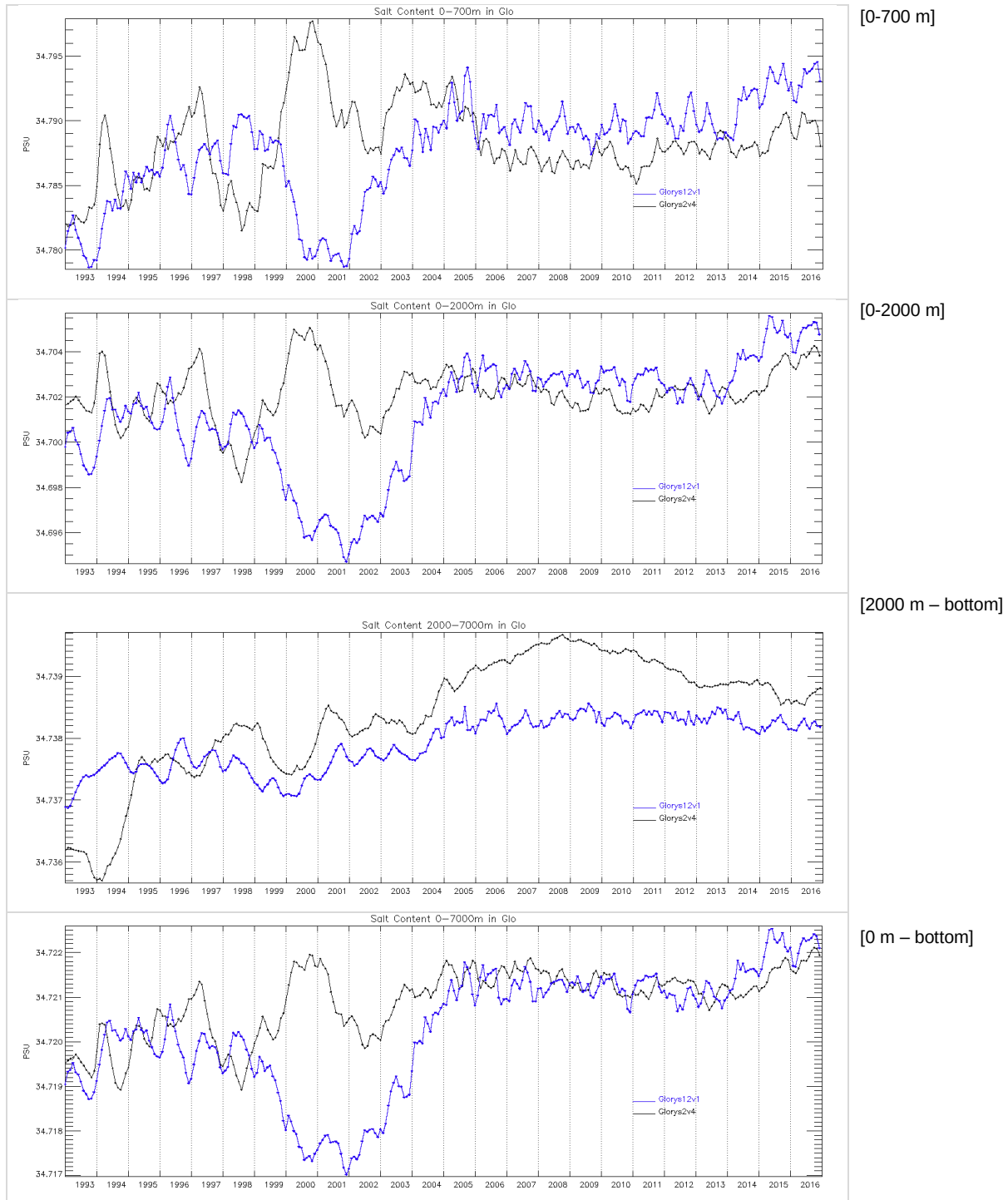


Fig. 2.5: Time series of global spatial averages of salinity (psu) in [0-700 m], [0-2000 m], [2000 m-bottom] and [0-bottom] layers for GLORYS2V4 (black) and GLORYS12V1 (blue).

IV.2.6 S-CLASS4-LAYER

For this validation we use the same product than in IV.1.7

IV.2.6.1 Timeseries

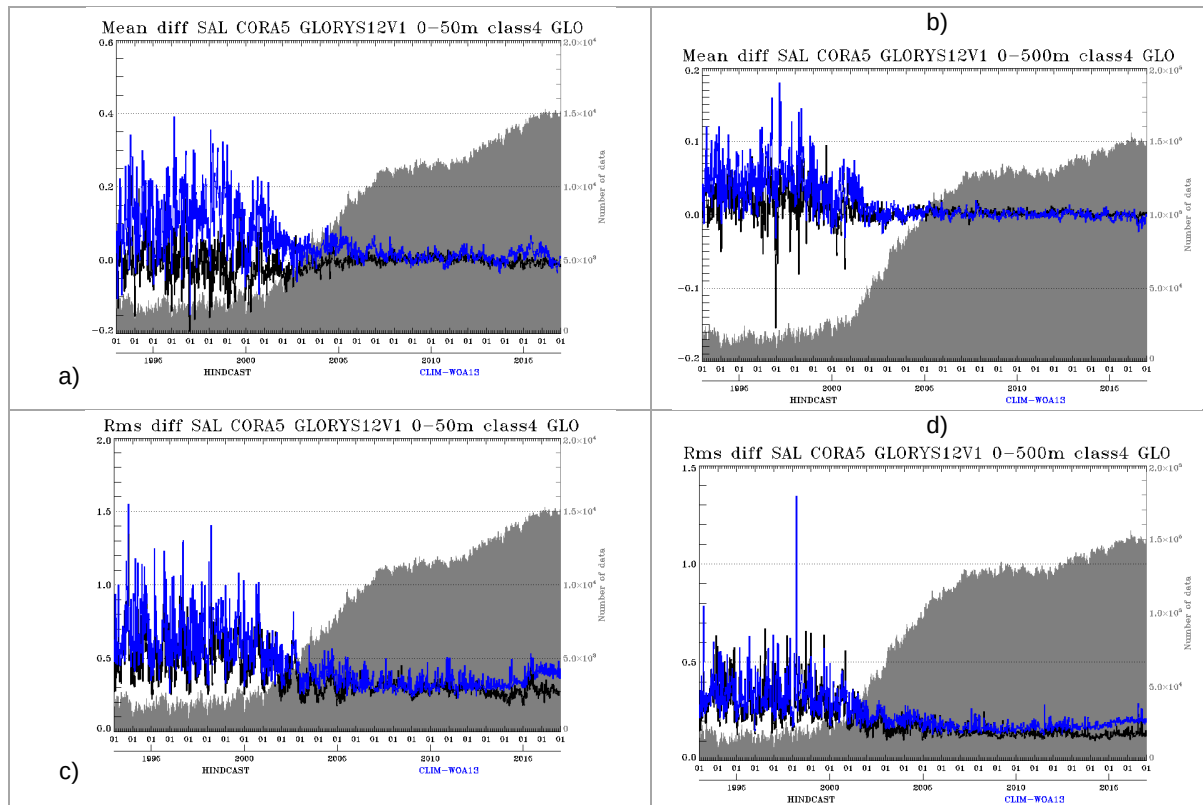


Figure 2.6.1: Timeseries (1993-2017) of (a, b) bias and (c,d) RMSD between model salinity and observations (in black) and temperature from climatology WOA13 and observations (in blue) for (a,c) the top 50 m and (b, d) the top 500 m. The grey shading on these graphics represents the number of data (cf., right y-axis for values)

The salinity scores are very stable starting in the early 2000's and from 2014 onward, the reanalysis clearly outperforms the climatology in terms of bias and RMSD with low RMSD in top 500 m. The bias is insignificant from the early 2000s onward for both the top 50m and top 500m.

IV.2.6.2 2D Maps

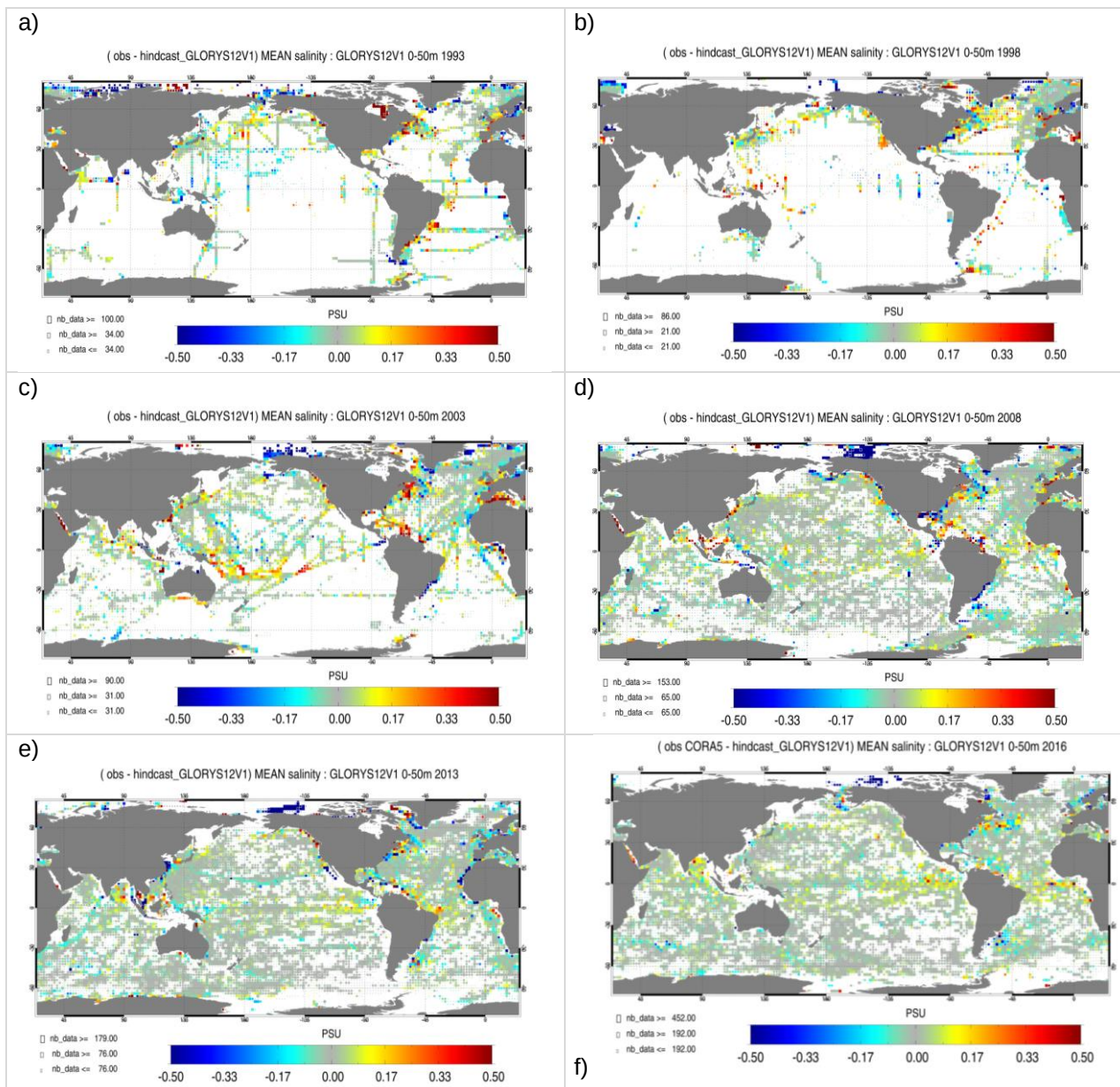


Figure 2.6.2: Salinity mean bias (observations-model) between the daily model averages and all available T/S observations from the Cora5.0 product for the years (a) 1993, (b) 1998, (c) 2003, (d) 2008, (e) 2013, and (f) 2016. Model outputs have been collocated with in situ sampling locations. *Data are averaged over the top 50m and the size of each pixel is proportional to the number of observations used to compute the difference in 2°x2° grid boxes.*

At the surface (Fig 2.6.2), the modelled salinities exhibit a fresh bias in the Eastern Tropical Pacific, the Tropical Atlantic, and in the Gulf of Bengal.

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	<table border="1"> <tr> <td>Ref</td><td>CMEMS-GLO-QUID-001-030</td></tr> <tr> <td>Date</td><td>November 2023</td></tr> <tr> <td>Issue</td><td>1.6</td></tr> </table>	Ref	CMEMS-GLO-QUID-001-030	Date	November 2023	Issue	1.6
Ref	CMEMS-GLO-QUID-001-030						
Date	November 2023						
Issue	1.6						

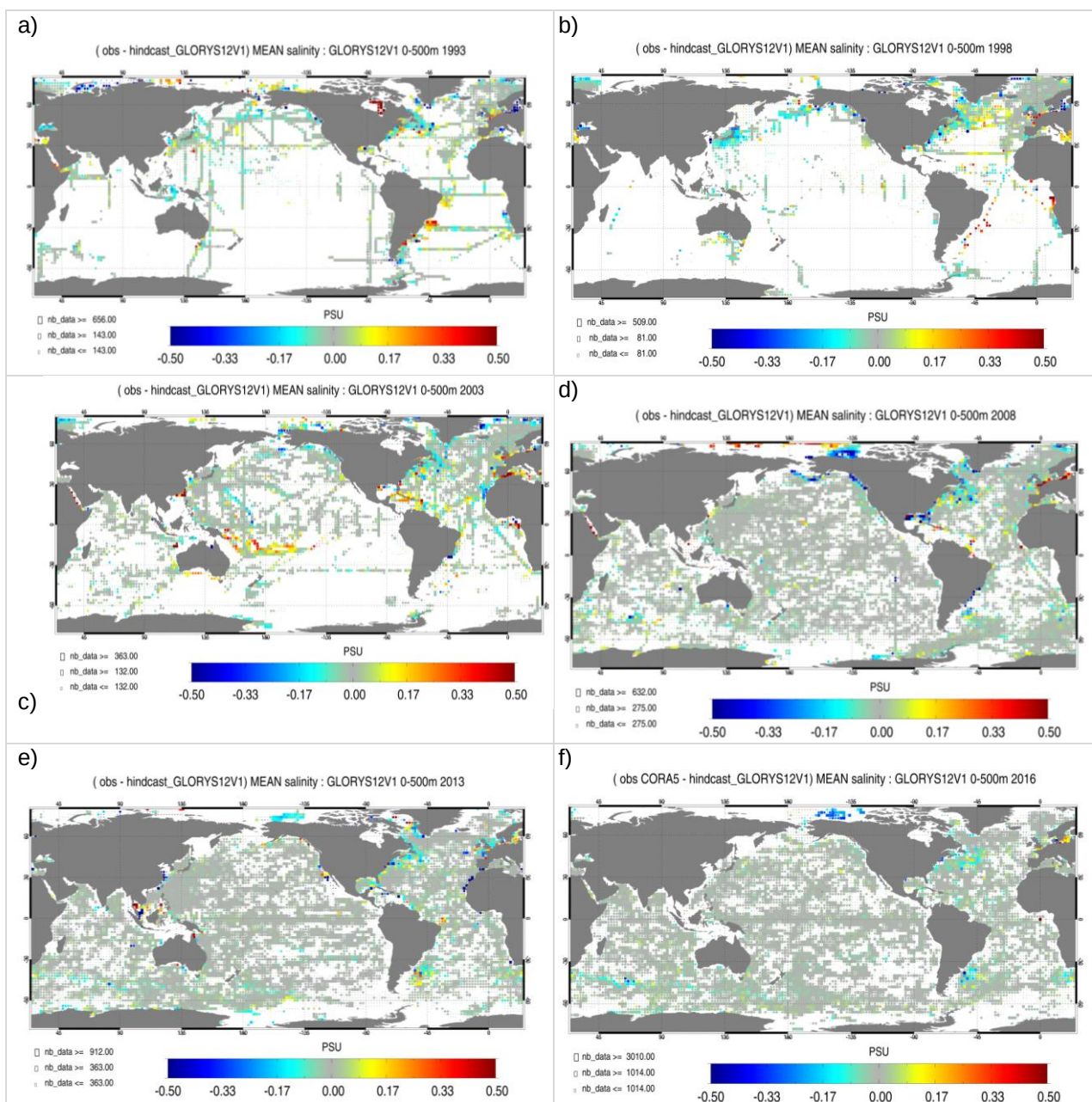


Figure 2.6.3 : As in Fig. 2.6.2 but averaged over the top 500 m.

If we average the top 500m, there is no significant bias (Fig. 2.6.3), except for a salty bias in the Beaufort Sea, probably due to a bad representation of the Summer Pacific Water. There is also a salty bias in some high mesoscale activity areas like the Gulf Stream and the Zapiola region.

IV.2.7 S-DA-INNO-STAT

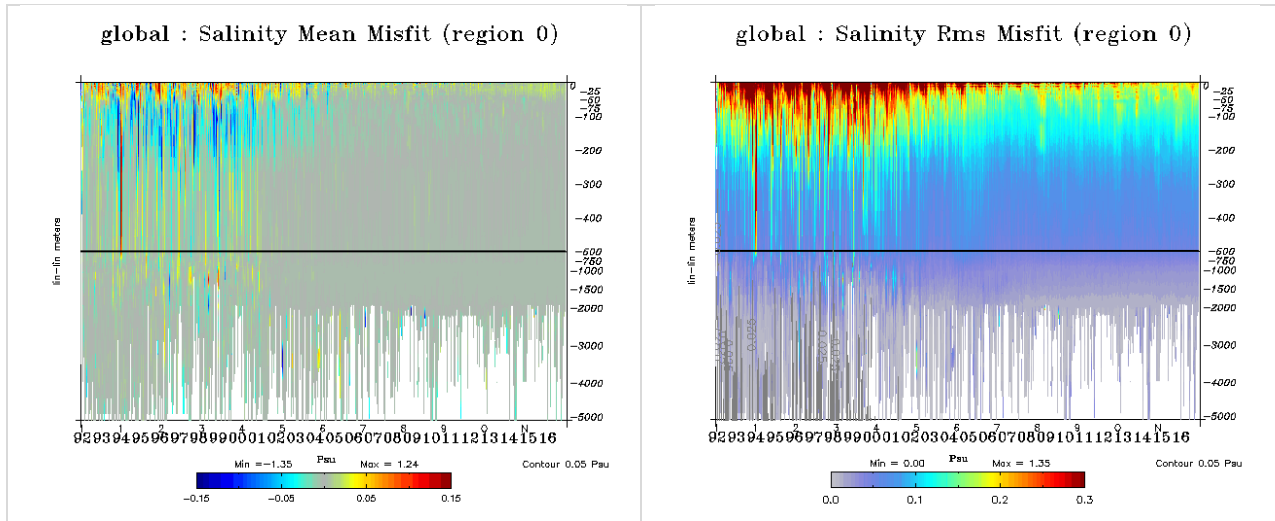


Fig 2.7: Time evolution of the domain average innovation in terms of bias (left) and RMSD (right) for CORA 4.1 in-situ salinity profiles. Covered time period: 1993 - 2016

As can be seen in Fig 2.7 general salty bias (less than 0.1 psu) is found in the [50 m-200 m] layer, mostly during the first half of the reanalysis. The fresh bias at the surface persists throughout the reanalysis (see also section IV.2.1). Both biases and RMSD against all the in-situ observations drastically decrease with time in accordance with the increasing observations network density.

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	<table border="1"> <tr> <td>Ref</td><td>CMEMS-GLO-QUID-001-030</td></tr> <tr> <td>Date</td><td>November 2023</td></tr> <tr> <td>Issue</td><td>1.6</td></tr> </table>	Ref	CMEMS-GLO-QUID-001-030	Date	November 2023	Issue	1.6
Ref	CMEMS-GLO-QUID-001-030						
Date	November 2023						
Issue	1.6						

IV.3 Currents

IV.3.1 *UV-CLASS1-15m_MEAN*

The main ocean currents are well reproduced by GLORYS12V1, as shown by the comparison with drifter velocity climatologies (Fig 3.1). Strong biases appear in the Western Tropical Pacific, where the eastward flowing equatorial current is not reproduced by GLORYS.

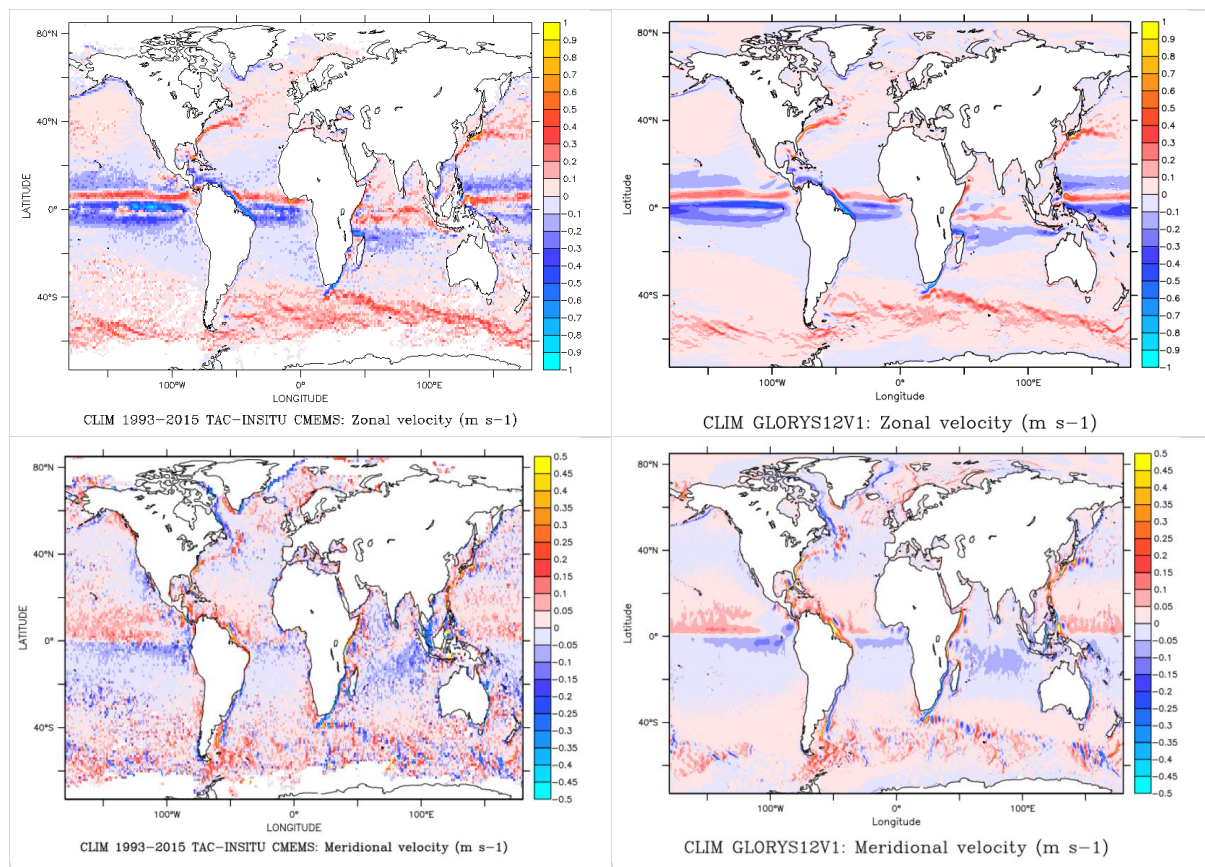


Fig. 3.1: Zonal (top) and meridional (bottom) 15 m depth current climatology for 1993-2014. NOAA AOML drifter-derived climatologies from INS TAC (left) and GLORYS12V1 (right).

IV.3.2 UV-CLASS2-MOORINGS

The vertical dynamics and structures of the upper layers in the tropics are well reproduced compared to mooring observations. However, the easterly surface current strength is overestimated in the Western Pacific (probably consistent with the near-surface biases in this area) while current strengths in the core of the Equatorial Under Current (EUC) is underestimated. RMSD are generally below 0.25 m s⁻¹ throughout the water column with a maximum in the lower part of the EUC. Correlations with equatorial moorings generally deteriorate with depth.

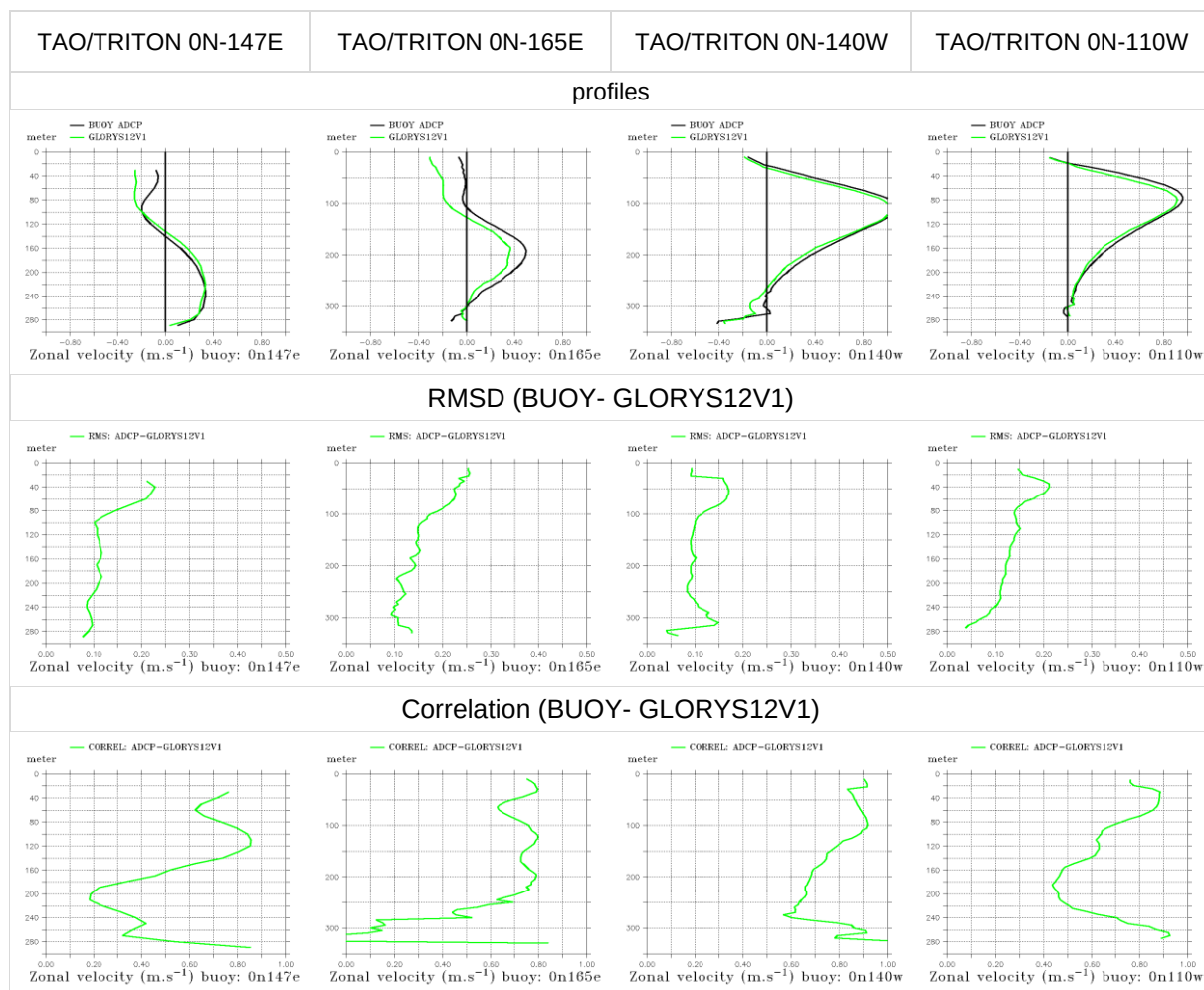


Fig. 3.2.1: Mean equatorial zonal velocity profiles (top), RMSD (middle), and correlations (bottom) estimated over the period 1993 – 2015 for observations (TAO/TRITON: 0N-140W, 0N-110W, 0N-165E, 0N-147E, PIRATA: 0N-23W, 0N-0E and RAMA: 0N-90E.) (black) and GLORYS12V1 (green).

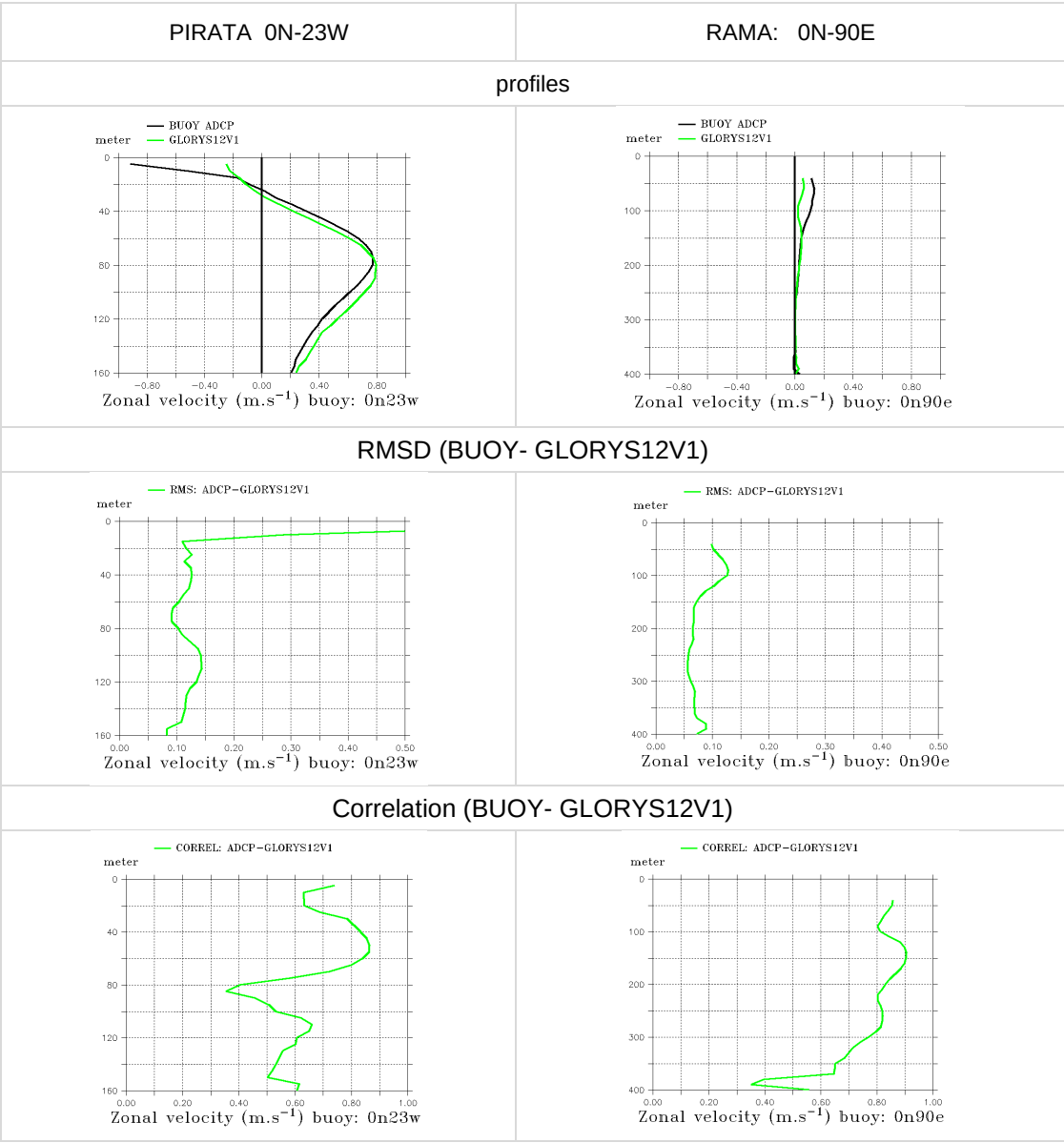


Fig. 3.2.2: Mean equatorial zonal velocity profiles (top), RMSD (middle), and correlations (bottom) estimated over the period 1993 – 2015 for observations (PIRATA:0N-23W and RAMA: 0N-90E) (black) and GLORYS12V1 (green).

IV.3.3 UV-CLASS2-MKE_1000m

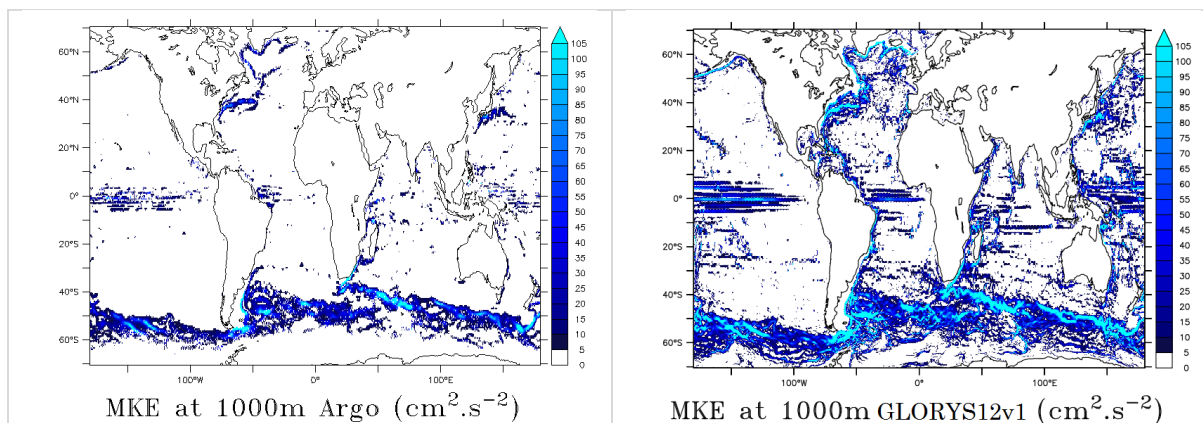


Fig. 3.3: Mean kinetic energy at 1000 m from the ANDRO Argo drift data base (left) and GLORYS12V1 (right). Covered time period: 2002-2009.

The modelled mean kinetic energy (MKE) at 1000 m is higher than suggested by ANDRO Argo drift data in all boundary currents and the ACC (Fig. 3.3). This possible overestimation of deep currents is still under investigation.

IV.4 Sea level

IV.4.1 SL-CLASS3-2DMEAN

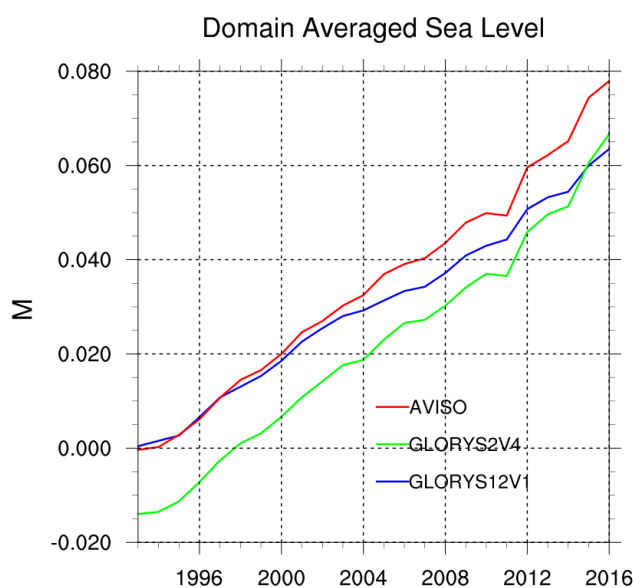


Fig. 4.1: Time series of domain-averaged annual mean sea level anomalies from Aviso SLA CCI observations (red), GLORYS12V1 (blue), and GLORYS2V4 (green). Covered time period: 1993-2016. The bias between AVISO data and GLORYS2V4 is due to the use of a different mean sea surface height and different reference periods (1993-2012 for altimetry and 1993-2016 for GLORYS).

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	Ref Date Issue	CMEMS-GLO-QUID-001-030 November 2023 1.6
---	-------------------------------	---

Linear trends SLA increases reach 3.27, 2.66, and 3.3 mm/year for SL-CCI, GLORYS12V1, and GLORYS2V4, respectively. However, with 2.66 mm/year, the globally averaged sea level trend is slightly underestimated compared to Aviso SLA CCI estimates of 3.27 mm/year. Nevertheless, regional trends are particularly well reproduced by GLORYS12 (not shown, see Lellouche et al., 2021). These trends are particularly strong at high northern latitudes, i.e., regions that are not well covered by altimetry measurements.

IV.4.2 *SL-CLASS4*

Not available yet for GLORYS12V1.

IV.4.3 *SL-DA-INNO_STAT*

GLORYS12V1 is very close to altimetry observations and has a good ability to describe sea level variability as indicated by a globally averaged RMSD of around 6cm (Fig. 4.3). However, a bias is diagnosed, which increases during the period, reaching around 1cm (the model is lower than observations) at the end of the period.

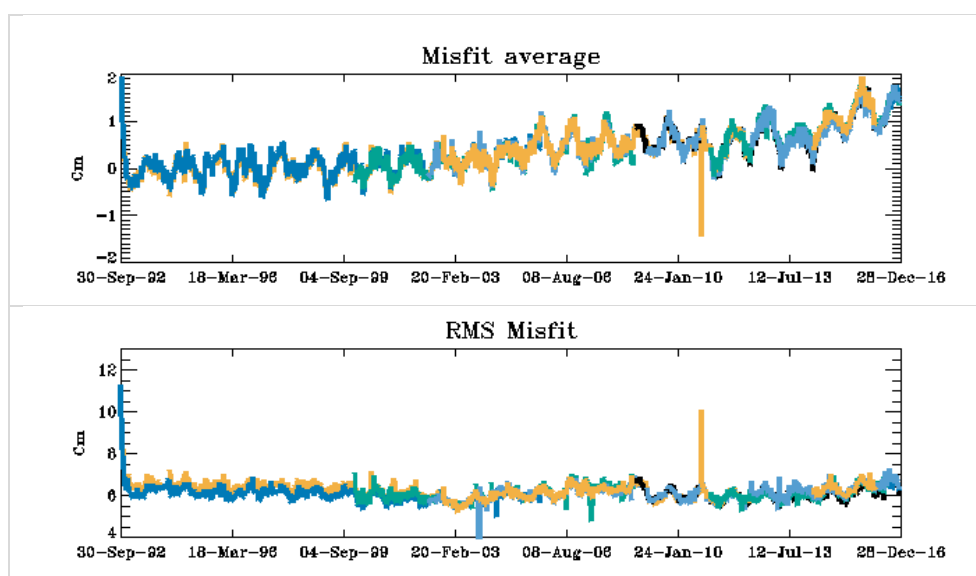


Fig. 4.3: Sea level innovation bias (top) and RMSD (bottom) over the whole domain (cm). Colours represent different altimetry satellites (ERS, T/P, Jason 1 & 2, Envisat, GFO, Saral/AltiKa, Cryosat2, HY2A). Covered time period: 1993 - 2016.

IV.5 Transport

IV.5.1 UV-CLASS3-VOL_TRANSP

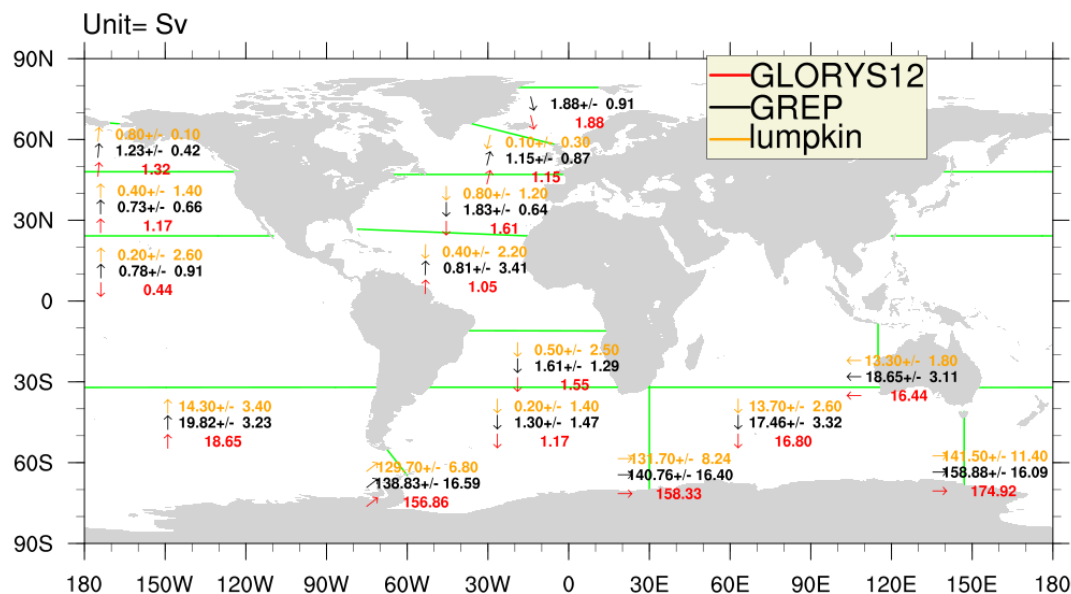


Fig 5.1: 1993-2016 mean volume transport (in Sv) from Lumpkin and Speer (2007) (orange), $\frac{1}{4}^\circ$ GLOBAL-REANALYSIS-PHY-001-026 ensemble product (black), and GLORYS12V1 reanalysis (red).

Assessing mean volume transports across different sections in the global ocean (Fig. 5.1), GLORYS12V1 transports are higher than in the $\frac{1}{4}^\circ$ ensemble product.

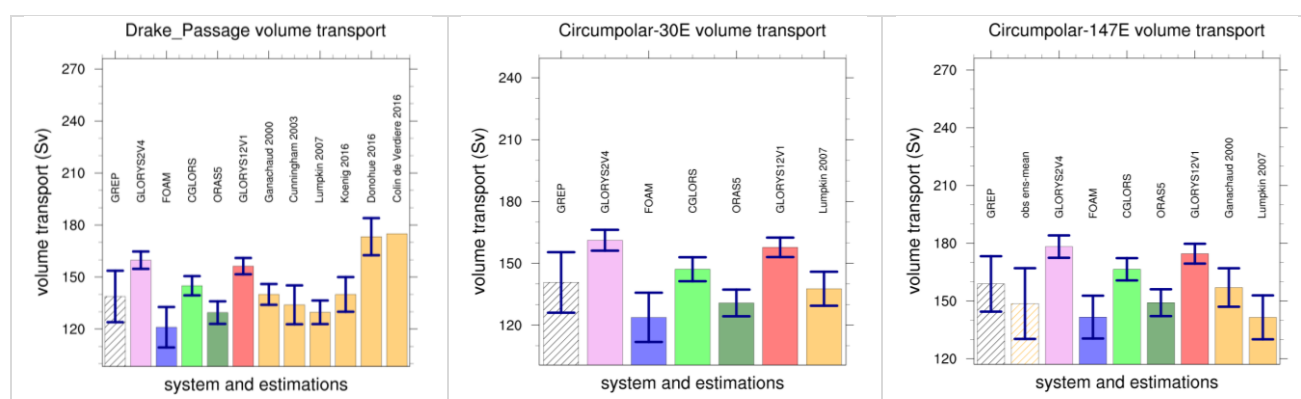


Fig 5.2: 1993-2016 mean volume transport (in Sv) estimates from observations (orange), $\frac{1}{4}^\circ$ GLOBAL-REANALYSIS-PHY-001-026 ensemble product (GREP) mean (black), individual GREP members (pink, blue, light green, dark green), and GLORYS12V1 (red); for Drake Passage (left), Cape of Good Hope-Antarctica (middle), and Tasmania-Antarctica sections.

In Figure 5.2, we present some ACC transports from GLORYS12 and other estimates. In the Drake Passage, recent estimates by Colin de Verdière and Ollitrault (2016) (175 Sv) and Donohue et al. (2017) (173.3 Sv) are higher and thus in better agreement with GLORYS12 compared to previous estimates by Ganachaud and Wunsch (2000) (140 Sv), Cunningham et al. (2003) (134 Sv), Lumpkin and Speer (2007) (130 Sv), and Koenig et al. (2014) (140 Sv).

IV.5.2 UV-CLASS3-AMOC

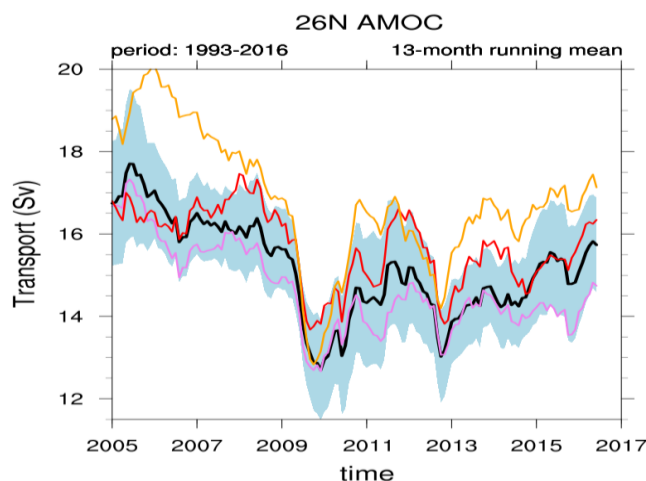


Fig 5.3: 1993-2017 meridional overturning current transport across 26°N for GREP ensemble reanalysis product (black line) and GREP spread (blue shaded area), GLORYS2V4 reanalysis at $\frac{1}{4}^\circ$ resolution (pink line), GLORYS12V1 reanalysis at $\frac{1}{12}^\circ$ resolution (red line) and RAPID observation-based estimation (orange).

GLORYS12V1 mean transport across 26°N by the meridional overturning current is 15.35 Sv for the 2009-2016 period and thus higher than the 14.06 Sv for the $\frac{1}{4}^\circ$ reanalysis GLORYS2V4 and also higher than 14.55 Sv of the mean of the multi-product at $\frac{1}{4}^\circ$ (REANALYSIS-PHY-001-031). The RAPID estimate (<http://www.rapid.ac.uk/>) is 16.01 Sv.

IV.5.3 T-CLASS3-MHT

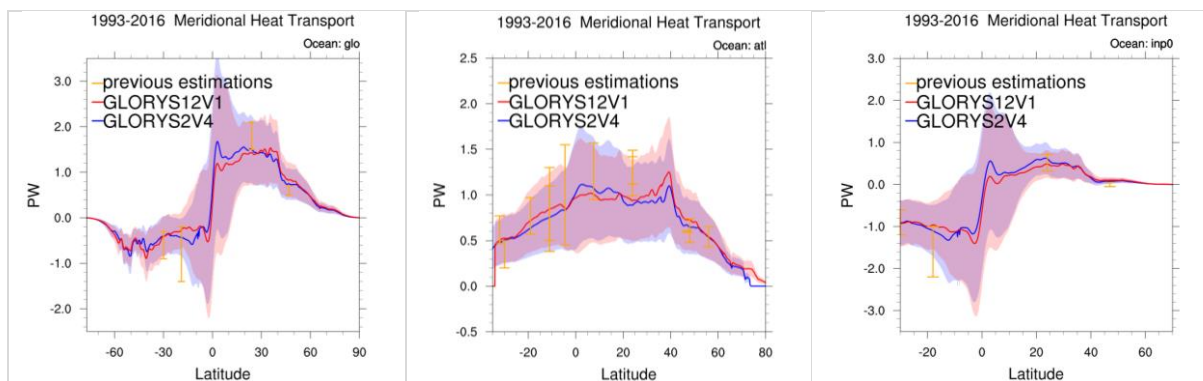


Fig 5.4: Meridional heat transport as a function of latitude. Left: Global Ocean, middle: Atlantic Ocean, right: Indian plus Pacific Oceans (units: Petawatts). Blue line and shaded area: GLORYS2V4 reanalysis at $\frac{1}{4}^\circ$ resolution; Red line and shaded area: GLORYS12V1 reanalysis at $\frac{1}{12}^\circ$ resolution; Orange: estimates provided by Ganachaud and Wunsch (2000) and Trenberth and Caron (2001) (black). Covered time period: 1993 - 2016.

Figure 5.4 presents the 1993-2016 meridional heat transport (MHT), computed from 5-day mean fields. We compare this $\frac{1}{12}^\circ$ reanalysis (red) to the $\frac{1}{4}^\circ$ reanalysis (blue).

The MHT of the $\frac{1}{12}^\circ$ reanalysis is of the same order of magnitude at higher latitudes compared to the $\frac{1}{4}^\circ$ reanalysis; At lower latitudes, the MHT of the $\frac{1}{12}^\circ$ reanalysis is weaker except in the Atlantic basin between 20°N and 40°N.

IV.6 Sea Ice

IV.6.1 *SI-CLASS1-CONC_MEAN*

The climatology of sea ice extent is well reproduced by GLORYS12 both in the Arctic (Fig. 6.1.1) and in the Antarctic (Fig. 6.1.2). The sea ice concentration is overestimated in the Antarctic in September.

The two major extrema during 1993-2016 are well reproduced (Fig 6.2)

Arctic

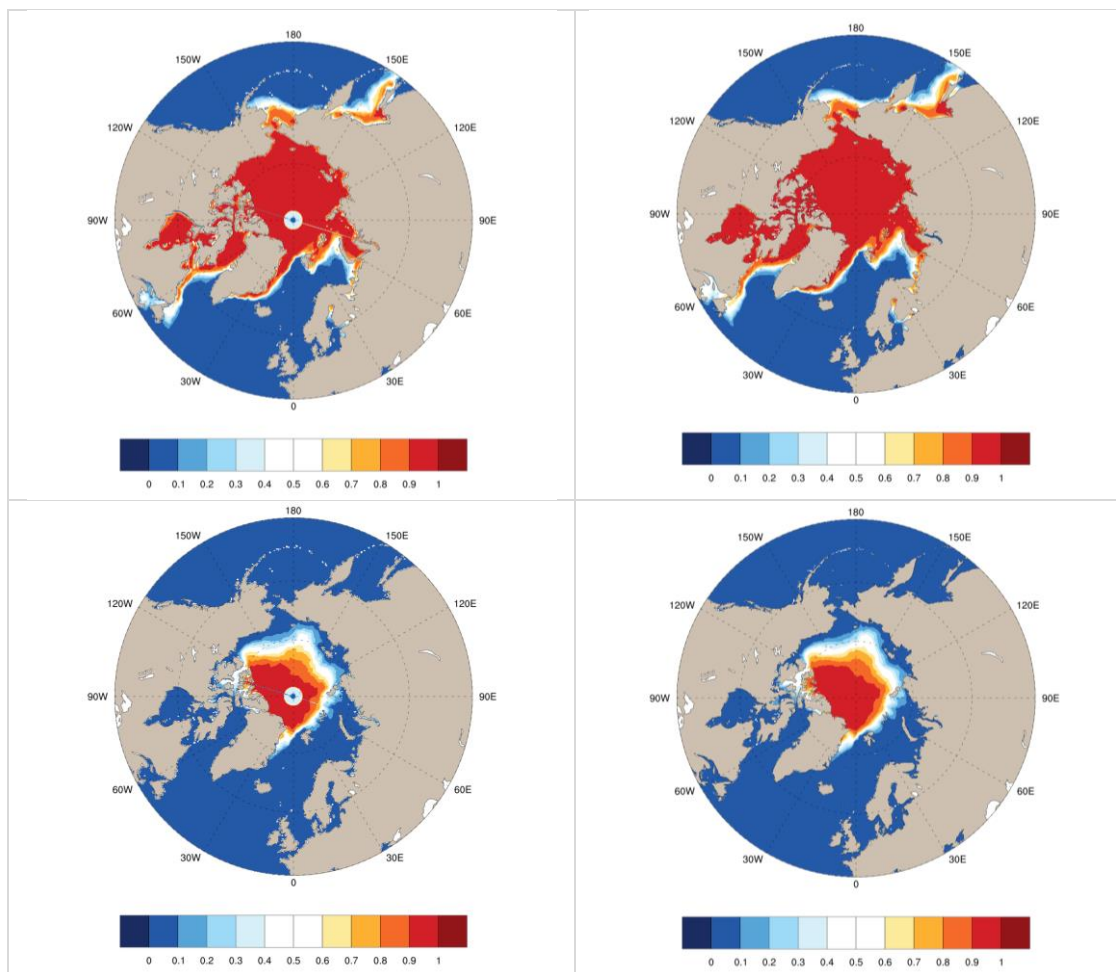


Fig 6.1.1: Arctic sea ice concentration climatologies (in %) (1993-2016) for March (top) and September (bottom). Left column: CERSAT observations; right: GLORYS12V1.

Antarctic

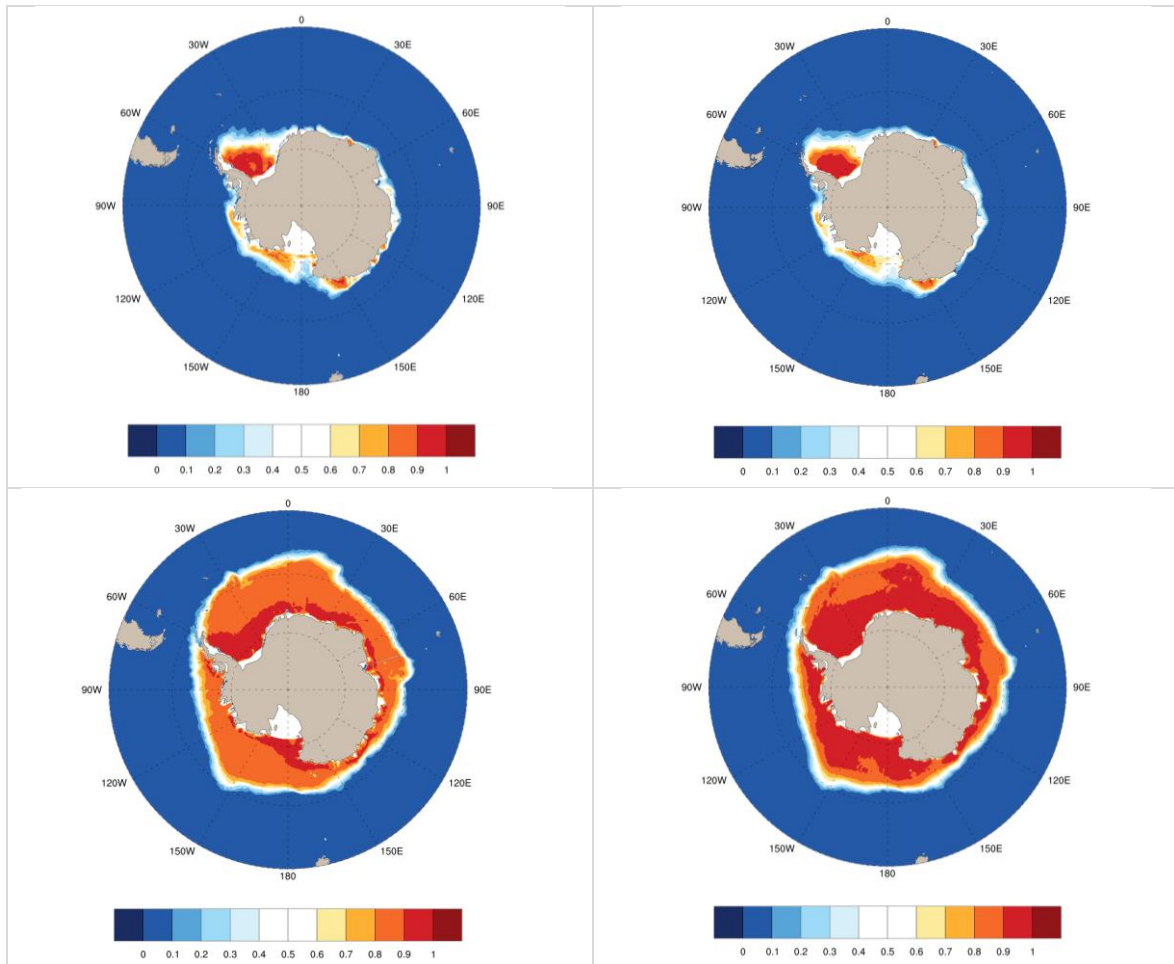


Fig 6.1.2: Antarctic sea ice concentration climatologies (in %) (1993-2016) for March (top) and September (bottom). Left column: CERSAT observations; right: GLORYS12V1.

IV.6.2 *SI-CLASS1-CONC_EXTREME*

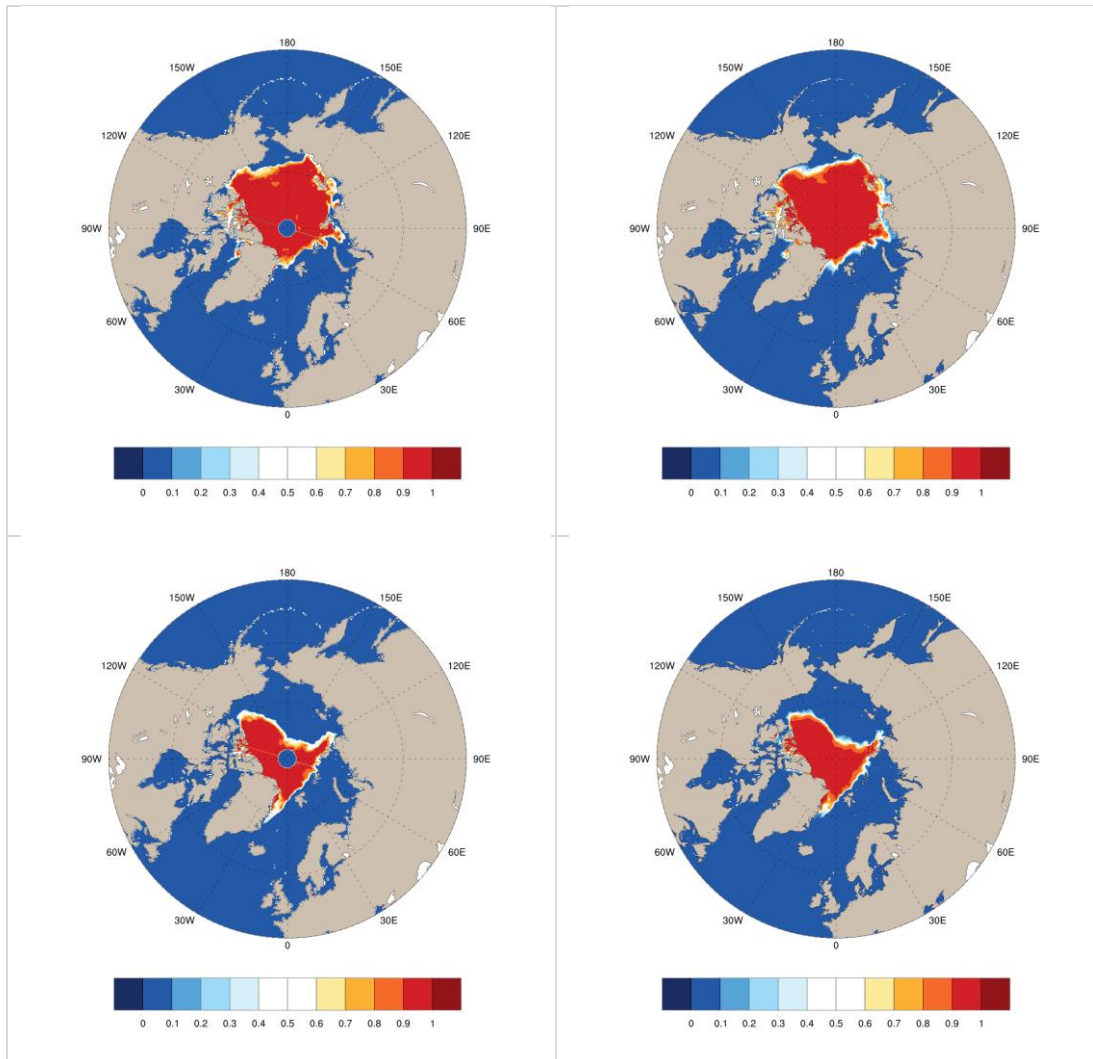


Fig 6.2: Arctic sea ice concentrations (in %) for September 1996 (top) and September 2007 (bottom). Left column: CERSAT observations; right: GLORYS12V1.

IV.6.3 SI-CLASS3-EXTENT

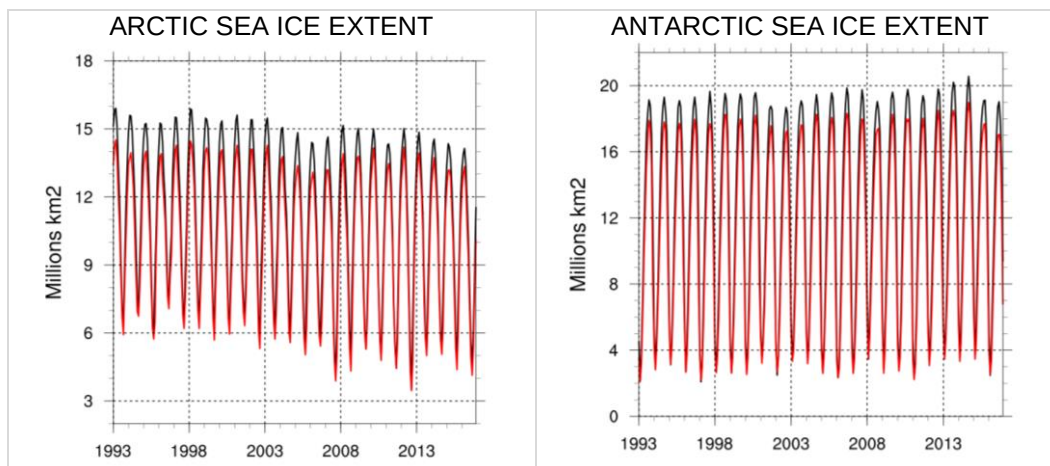


Fig 6.3: Integrated sea ice extent in the Arctic (left) and Antarctic (right) (units: 10^6 km^2). CERSAT observations (red) and GLORYS12V1 (black). Covered time period: 1993-2016.

Both Arctic sea ice extent (Figs. 6.3 and 6.4) and volume (Figs. 6.5 and 6.6) show a general respective decrease of $-85,400 \text{ km}^2/\text{year}$ and $-46,200 \text{ km}^3/\text{year}$ for the 1993-2016 period. Winter sea ice extent maxima in both hemispheres are overestimated compared to the assimilated CERSAT data. Although the year 2016 exhibits an unusually strong negative anomaly, the Antarctic sea ice extent still exhibits a positive trend of $56,300 \text{ km}^2/\text{year}$ for the 1993-2016 period (Fig 6.4).

IV.6.4 SI-CLASS3-EXTENT_ANO

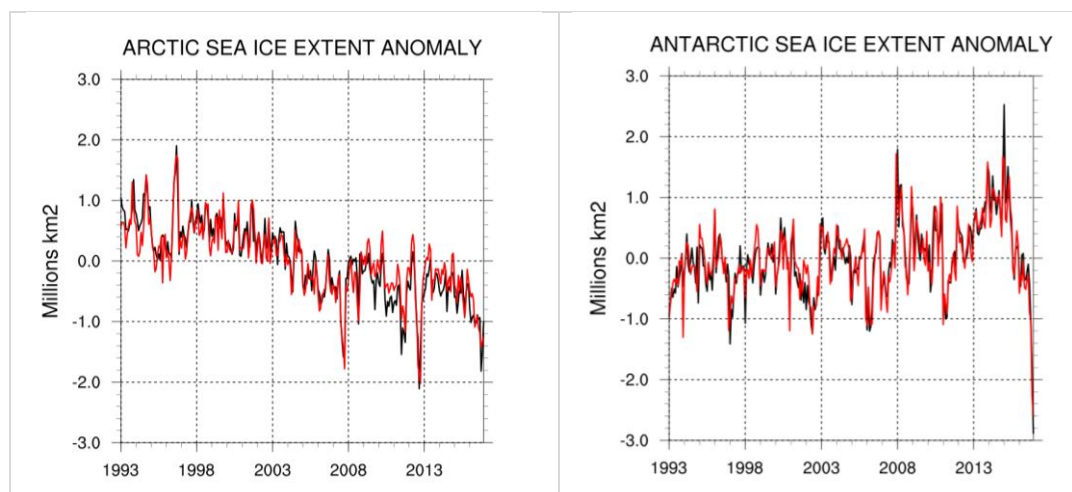


Fig 6.4: Integrated sea ice extent interannual anomalies in the Arctic (left) and Antarctic (right) (units: 10^6 km^2). CERSAT observations (red) and GLORYS12V1 (black). The linear correlation (not de-trended) between GLORYS12V1 and CERSAT observation is 0.93 and 0.91 for the Arctic and Antarctic, respectively. Covered time period: 1993-2016.

IV.6.5 SI-CLASS3-VOL

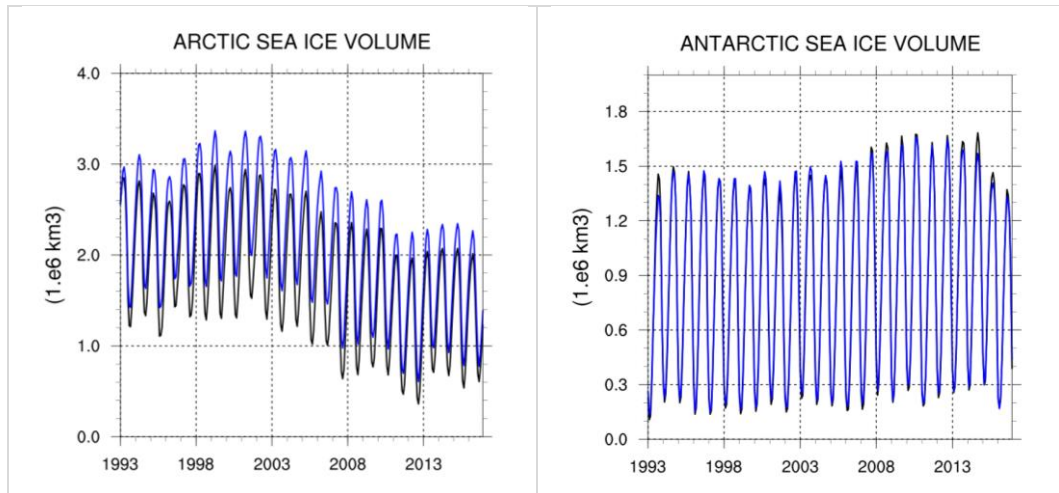


Fig 6.5: Integrated sea ice volume in the Arctic (left) and Antarctic (right) (units: 10^6 km^3) for GLORYS12V1 (black) and GLORYS2V4 (blue). Covered time period: 1993-2016.

IV.6.6 SI-CLASS3-VOL_ANO

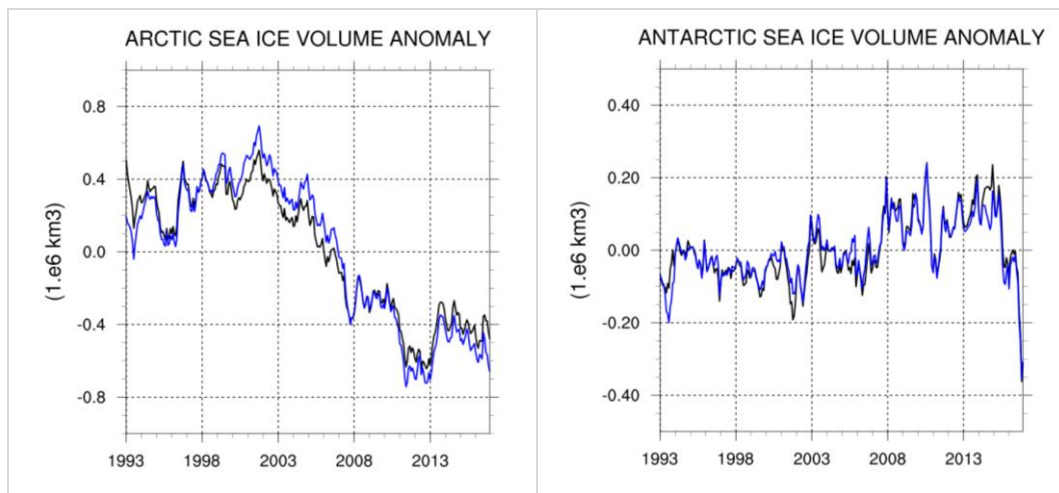


Fig 6.6: Integrated sea ice volume interannual anomalies in the Arctic (left) and Antarctic (right) (units: 10^6 km^3) for GLORYS12V1 (black) and GLORYS2V4 (blue). Covered time period: 1993-2016.

IV.6.7 SI-CONC-DA-INNO-STAT

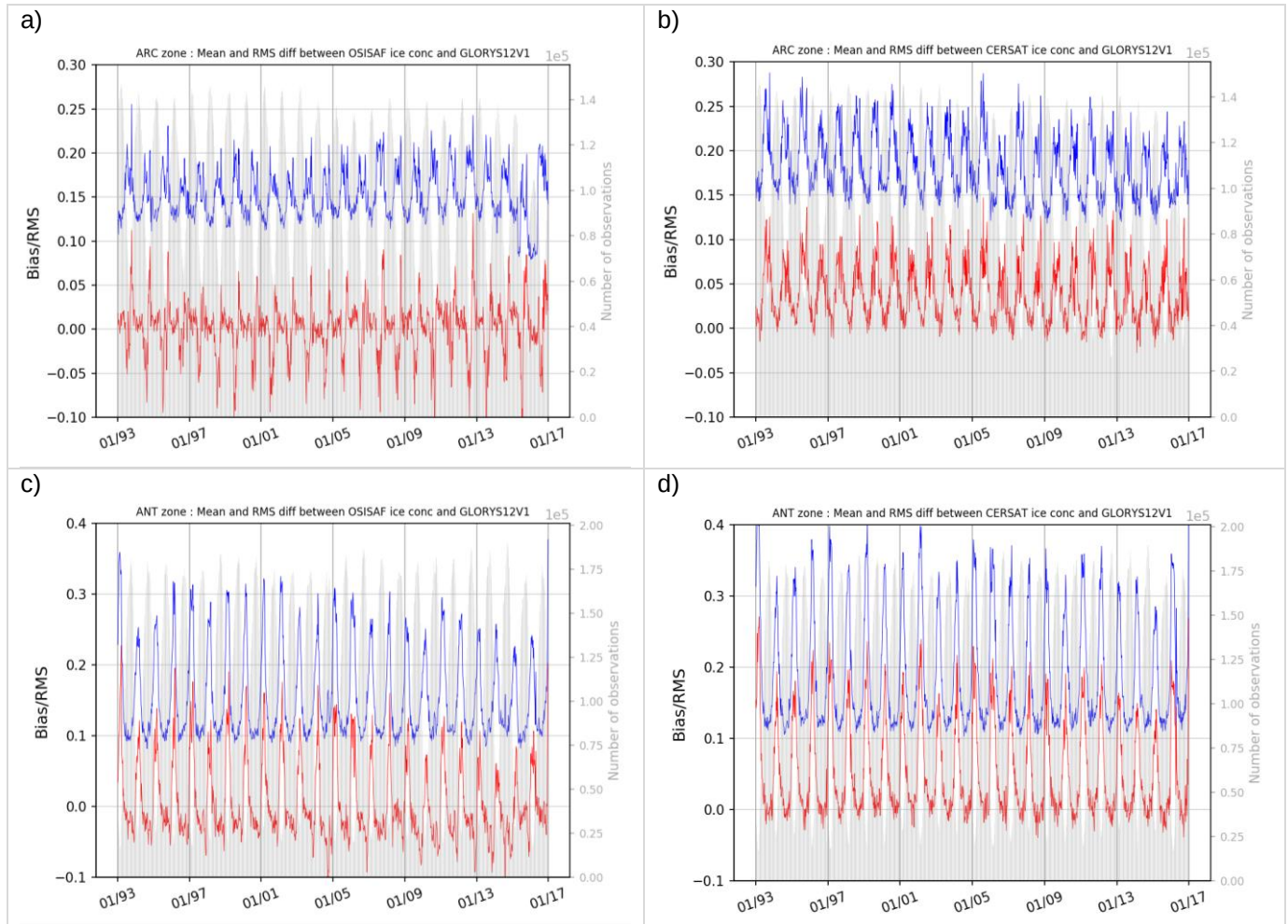


Figure 6.7: Bias (observation-analysis) (red) and RMSD (blue) of sea ice concentration (0 means no ice, 1 means 100% ice cover) in the Arctic Ocean (a and b) and Antarctic Ocean (c and d). The assimilated observations are the sea ice concentrations from CERSAT (a and c) and the independent dataset OSI TAC (b and d). The number of observations is shown as grey bars (right axes).

Due to sea ice concentration assimilation, sea ice extent variability (seasonal cycles, interannual variability, and trends) is well reproduced in GLORYS12V1.

RMSD and biases with respect to the assimilated ice concentration dataset are stable throughout the validation period (Fig 6.7). The maximum RMSD in the Arctic reaches 0.25 in summer and 0.12 in winter. The results are generally the same with the independent dataset but somehow higher. Positive differences between the observations and the model during spring and summer show an excess of ice melting in GLORYS12V1. During winter, the differences are positive indicating an underestimation of the spread of ice extent. Results are similar in the Antarctic although RMSD are high and reach up to 0.4 (for sea ice concentration).

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	<table> <tr> <td>Ref</td><td>CMEMS-GLO-QUID-001-030</td></tr> <tr> <td>Date</td><td>November 2023</td></tr> <tr> <td>Issue</td><td>1.6</td></tr> </table>	Ref	CMEMS-GLO-QUID-001-030	Date	November 2023	Issue	1.6
Ref	CMEMS-GLO-QUID-001-030						
Date	November 2023						
Issue	1.6						

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

Since the ERA-interim atmospheric reanalysis was discontinued in 2019, we used ERA5 forcing fields starting from January 2019 (cf., Section II.1).

From July 2021 onward, GLORYS12V1 production switched to interim mode, which means that it is produced by assimilating the following NRT observational data:

- INSITU_GLO_PHY_TSASSIM_DISCRETE_NRT_013_047 for in situ;
- SEALEVEL_GLO_PHY_L3_NRT_OBSERVATIONS_008_044 for altimetry;
- v2.1 REYNOLDS for SST;
- CERSAT from Ifremer for sea ice concentration.

In the future, GLORYS12V1 will be routinely extended by one month each month to guarantee data availability close to real time.

Initial diagnostics show that GLORYS12V1 produces a realistic representation of the ocean state, with a realistic temperature/salinity RMSD (Fig 7.1). However, it is important to note that the observed warming, particularly at depths of around 100m, as well as the SLA bias (Lellouche et al. 2021) have continued to intensify over recent years (Figs. 7.1 and 7.2).

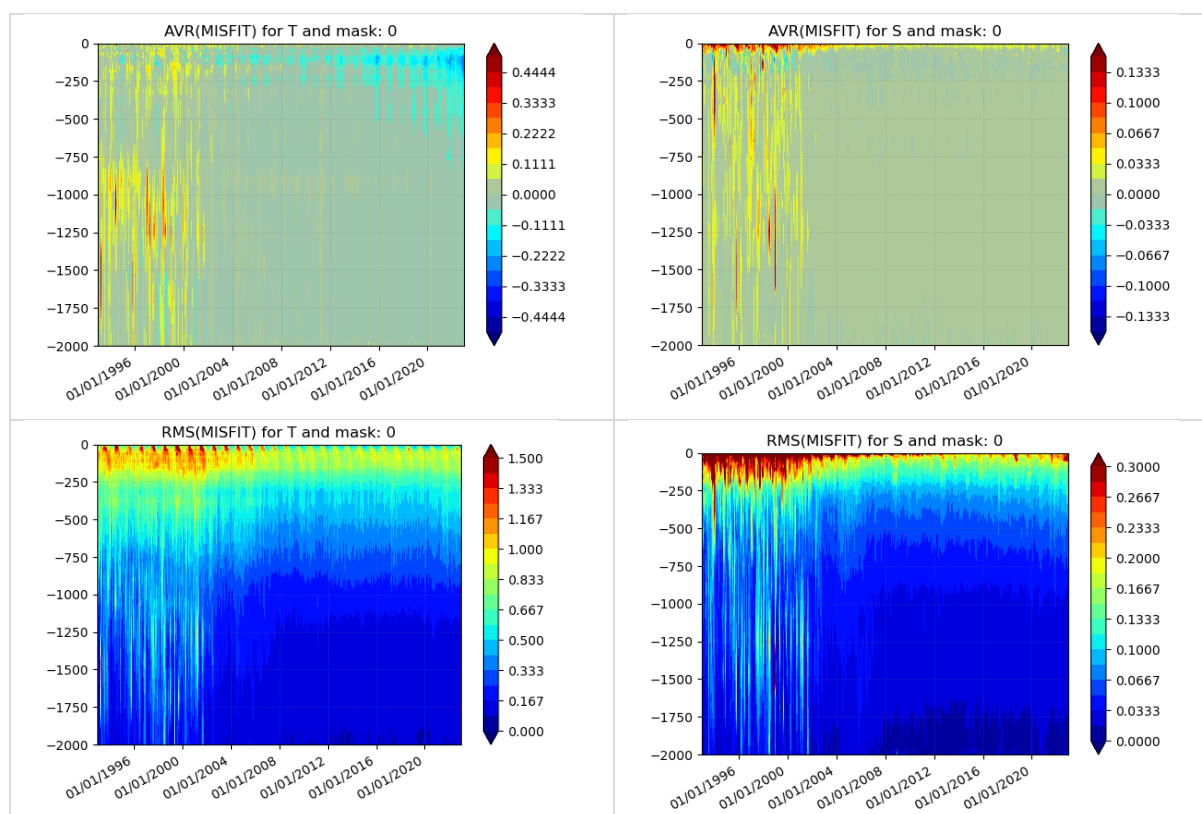


Figure 7.1: Bias (observation-analysis) (top) and RMSD (bottom) for temperature (left) and salinity (right) of GLORYS12V1 from January 1993 to July 2023.

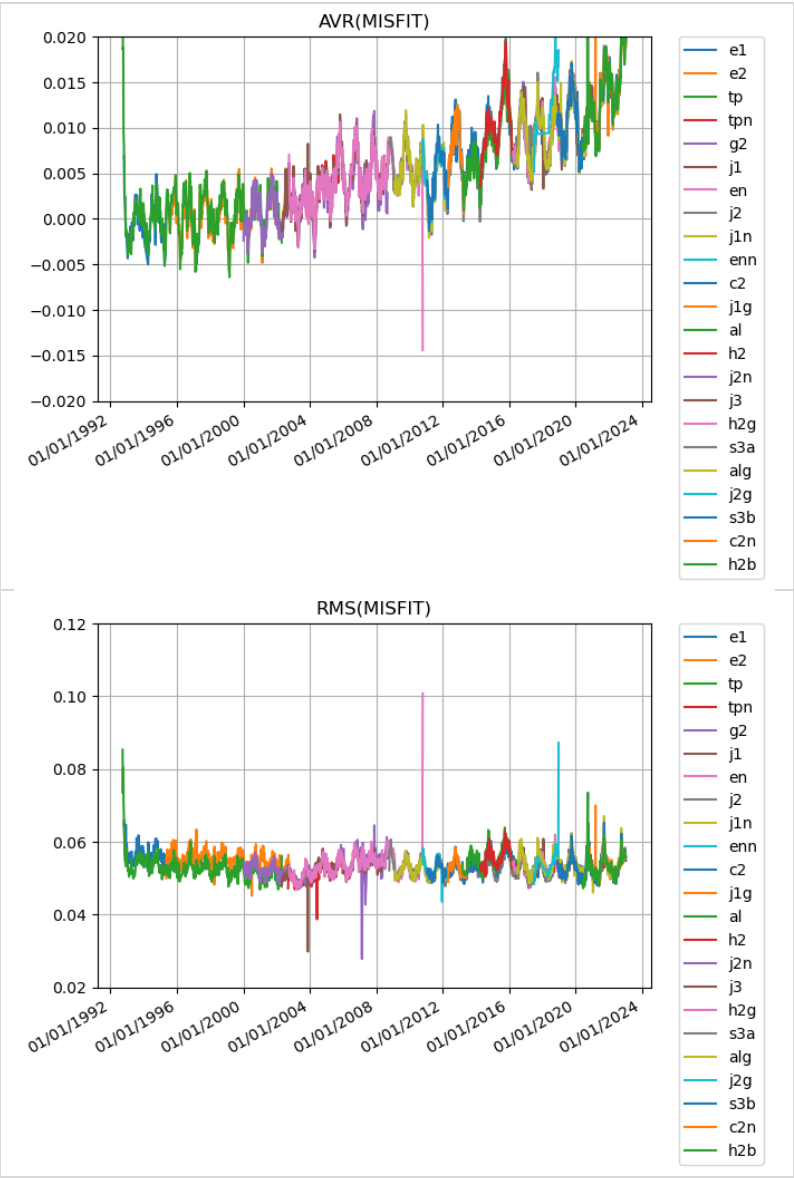


Figure 7.2: SLA bias (top) and RMSD (bottom) with respect to different altimetry missions of GLORYS12V1 from January 1993 to July 2023.

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	Ref Date Issue	CMEMS-GLO-QUID-001-030 November 2023 1.6
--	----------------------	--

VI QUALITY CHANGES SINCE PREVIOUS VERSION

Quality Information Document For products GLOBAL_REANALYSIS_PHY_001_030	Ref Date Issue	CMEMS-GLO-QUID-001-030 November 2023 1.6
---	-------------------------------	---

VII REFERENCES

- Colin de Verdière, A., & Ollitrault, M. (2016). A direct determination of the World Ocean barotropic circulation. *Journal of Physical Oceanography*, 46(1), 255-273.
- Cunningham, S. A., S. G. Alderson, B. A. King, and M. A. Brandon, Transport and variability of the Antarctic Circumpolar Current in Drake Passage, *J. Geophys. Res.*, 108(C5), 8084, doi:10.1029/2001JC001147, 2003.
- Donohue, K. A., Tracey, K. L., Watts, D. R., Chidichimo, M. P., & Chereskin, T. K. (2016). Mean antarctic circumpolar current transport measured in drake passage. *Geophysical Research Letters*, 43(22), 11-760.
- Drevillon et al (2018) Report on validation guidelines for ocean reanalyses of the Copernicus Marine Environment Monitoring Service, CMEMS-RAN-ScV-common-plan
- Ganachaud, A., & Wunsch, C. (2000). Improved estimates of global ocean circulation, heat transport and mixing from hydrographic data. *Nature*, 408(6811), 453-457.
- Gouretski, V.V. and Koltermann, K.P.: Woce global hydrographic climatology. Technical Report 35/2004, Berichte des Bundesamtes fur Seeschifffahrt und Hydrographie, 2004.
- Hilburn, K. A., (2009), [The passive microwave water cycle product](#), report number 072409, Remote Sensing Systems, 30 pp.
- Koenig, Z., C. Provost, R. Ferrari, N. Sennechael, and M.-H. Rio (2014), Volume transport of the Antarctic Circumpolar Current: Production and validation of a 20 year long time series obtained from in situ and satellite observations, *J. Geophys. Res. Oceans*, 119, 5407–5433, doi:10.1002/2014JC009966.
- Koenig, Z., Provost, C., Park, Y. H., Ferrari, R., & Sennéchael, N. (2016). Anatomy of the Antarctic Circumpolar Current volume transports through Drake Passage. *Journal of Geophysical Research: Oceans*, 121(4), 2572-2595.
- Lellouche, J.-M., Greiner, E., Le Galloudec, O., Garric, G., Regnier, C., Drevillon, M., et al. (2018). Recent updates on the copernicus marine service global ocean monitoring and forecasting real-time 1/12° high resolution system. *Ocean Sci.* 14, 1093–1126. doi: 10.5194/os-14-1093-2018
- Lellouche, J.-M., Eric, G., Romain, B. B., Gilles, G., Angelique, M., Marie, D., et al. (2021). The copernicus global 1/12° oceanic and sea ice GLORYS12 reanalysis. *Frontiers of Earth Science*, 9, 698876. <https://doi.org/10.3389/feart.2021.698876>
- Lumpkin, R., & Speer, K. (2007). Global ocean meridional overturning. *Journal of Physical Oceanography*, 37(10), 2550-2562.
- Moat, B. I., Smeed, D. A., Frajka-Williams, E., Desbruyères, D. G., Beaulieu, C., Johns, W. E., Rayner, D., Sanchez-Franks, A., Baringer, M. O., Volkov, D., Jackson, L. C., and Bryden, H. L.: Pending recovery in the strength of the meridional overturning circulation at 26° N, *Ocean Sci.*, 16, 863–874, <https://doi.org/10.5194/os-16-863-2020>, 2020.
- Trenberth, K. E., & Caron, J. M. (2001). Estimates of meridional atmosphere and ocean heat transports. *Journal of Climate*, 14(16), 3433-3443.